The Value of Outside Options: Measurement, Impact and Inequality

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How do other employers shape a worker's pay? Using Portuguese matched employer–employee data, I identify labor markets from job mobility patterns and measure how the pay quality of alternative employers affects wages. Across markets, a 10% increase in outside options pay quality is associated with 3% higher wages. Within the same market and year, however, the same increase is associated with 1% lower current wages. This markdown is strongest among new hires and early-career workers, fades with tenure, and is followed by faster wage growth among workers who stay. Together, these patterns support a dynamic compensation story: valuable outside options shift compensation from current pay toward future progression. Despite shaping individual wages, this channel contributes little to aggregate wage inequality. Outside options nonetheless matter for inequality through a different channel, as market-level pay conditions account for about 15% of wage variance in Portugal.

Keywords: Outside options; wage determination; community detection; peer effects; wage inequality.

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1. Introduction

Do wages in other firms matter for my own wage? Understanding the extent to which individual wages depend on the pay practices of competing firms is a crucial step toward understanding how wages are set and why persistent wage inequality arises. If wages react to the pay practices of other firms, labor markets become not just a collection of bilateral employer–employee relationships, but a network in which firms' strategies are interdependent. This enhances the role of labor market power, as firms can leverage this to mark down wages relative to productivity. Exploring this interdependence can shed light on the sources of wage dispersion, the mechanisms that sustain inequality, and the role of competition in shaping pay.

The idea that outside options influence wage setting has a long theoretical tradition (Farber 1986; Pissarides 2000). Yet, translating these insights into empirical evidence has proven difficult. A central challenge lies in identifying a worker's relevant labor market (i.e., who their true competitors and alternatives are) and in measuring the degree of wage interdependence across firms. Without accounting for the networked structure of the labor market and the possibility that firms react to one another's wage policies, empirical analyses risk drawing biased conclusions about how wages are formed and how inequality emerges.

In this paper, I estimate the impact of other firms' pay policies on a worker's wage, conditional on observed and unobserved characteristics, using matched employer-employee data covering the universe of Portuguese private firms from 2010 to 2021. Drawing from the peer effects literature, I construct a non-linear least squares estimator that enables the estimation of worker, firm and market unobserved heterogeneity parameters while simultaneously recovering how the quality of other firms in the labor market affects wages. I summarize the pay quality of alternative employment opportunities with a leave-out average of firm wage premia within a labor market, weighted by firm-market employment shares.

In a benchmark model that leverages cross-market variation, I find that a 10% increase in the average quality of outside firms (as given by their firm fixed effects) is associated with a 2.8% increase in real hourly wages. However, when controlling for common shocks to the market that affect outside options simultaneously for all workers and exploring within-market-year variation, the sign flips, with this elasticity turning to –0.1. The positive benchmark elasticity and negative conditional estimates are not, however, competing answers to the same question. The first describes how wages move with the overall quality of the surrounding market; the second isolates the relationship with a worker's relative exposure to alternative employers within that market and year.

Search models with on-the-job offers, bargaining, and renegotiation provide the theoretical foundation for interpreting this elasticity (Postel-Vinay and Robin 2002; Cahuc,

Postel-Vinay, and Robin 2006). In these models, a worker who receives an outside offer may use it to renegotiate with her current employer or move to the offering firm. Better alternatives raise the value of search and strengthen the worker's future bargaining position. At the same time, a firm that anticipates this future competition may backload compensation, paying less at the beginning of the relationship and raising wages when outside offers arrive. The prospect of better future opportunities can therefore produce a negative relationship between alternative-employer pay quality and current wages even when those opportunities improve the value of the job. Learning and the accumulation of portable skills reinforce this logic: if the current job makes the worker more valuable to high-paying firms, part of that future return can be priced into a lower starting wage. The empirical analysis tests the wage profile, mobility, and promotion implications of this dynamic-compensation mechanism.

To start the analysis, I classify jobs into labor markets. Using the approach of Nimczik (2020), I borrow from the network studies literature—more specifically the work on community detection—and assume that the labor market network follows a Stochastic Block Model structure (Holland, Laskey, and Leinhardt 1983; Karrer and Newman 2011). In this setting, jobs are split into several blocks (markets), in which workers have a higher mobility probability. The detection of these blocks is done through the minimization of the likelihood function for the network-generating process. This procedure captures markets by clustering jobs with large mobility patterns between them. This yields hundreds of labor markets, in which I assume that workers predominantly search for jobs.

With the labor markets defined, I establish that the leave-out average of firm fixed effects contains information about the firms workers can reach. Among workers who change employers within the same market, this leave-out average term strongly predicts the premium of the destination firm. This reveals that workers surrounded by higher-premium employers therefore move to higher-premium firms, which validates the leave-out average as a measure of the pay quality of relevant alternatives rather than an abstract feature of market composition.

Then, I employ techniques from the peer effects literature that estimate the impact of peer quality on a given outcome in the presence of high-dimensional unobserved heterogeneity (Arcidiacono et al. 2012):

$$w_{ifmt} = \mathbf{x}'_{ifmt}\boldsymbol{\beta} + \alpha_i + \zeta_f + \theta_{mt} + \gamma\overline{\zeta}_{-fmt} + \varepsilon_{ifmt}. \tag{1}$$

Here, α_i captures the worker's permanent wage component, ζ_f the premium paid by her current firm, and θ_{mt} wage determinants shared by workers in market m and year t. The final term, $\overline{\zeta}_{-fmt}$, is the employment-weighted average premium of all other firms in the worker's market and occupation. The elasticity γ measures wage interdependence across workplaces: it asks whether current wages vary with the pay quality of relevant alternative

employers after accounting for who the worker is, where she currently works, and the wage conditions common to her market. This formulation circumvents Manski (1993)'s reflection problem, as there is no potential for reverse causality between the observed wage of individual i and the pay policies of other firms after controlling for worker and firm unobserved heterogeneity.

Because market-by-year fixed effects absorb unobserved shocks common to all firms within a market-year, the elasticity is identified by variation in market structure, which I specify as differences and changes in firms' employment shares. The intuition is that if employment is distributed evenly across firms within a market, each firm is equally likely to be encountered by a searching worker, leaving no cross-firm variation in the expected value of outside employment. When firm sizes differ, however, workers are more likely to encounter larger firms. Consequently, heterogeneity in firm size across markets and over time generates the variation used to identify the elasticity of interest. I find that, without market-year fixed effects, a 10% increase in the average quality of outside firms leads to an increase in wages of close to 3%. However, when controlling for common market-level shocks through the fixed effects, this elasticity drops to −0.1. This is because, without a measure that absorbs common shocks to outside options, the leave-out average of firm effects also captures local supply and demand shocks, which affect the share of the outside option value that workers receive. After controlling for market-year effects, however, what remains are idiosyncratic shocks that are either worker-firm specific or truly outside the firm.

I then show that this current-pay markdown is part of a steeper wage and career profile. The negative elasticity is concentrated near hiring. During the first year with an employer, a 10% increase in alternative-employer pay quality is associated with wages that are 1.5% lower. This elasticity becomes 0.9% for workers who have between one and three years of tenure and becomes close to zero between three and ten years. I also find that future renegotiation, promotions, and separations are all positively associated with higher outside option quality. The impact of outside opportunities is also much larger for early-career workers. The markdown is therefore largest when the future value of renegotiation, promotion, and job mobility is greatest, and it largely unwinds as the employment relationship matures.

How does this affect inequality? The work of Abowd, Kramarz, and Margolis (1999) (AKM) is influential in this regard, as it proposed a statistical model that attributes a large share of wage disparities to differences in worker ability, firm-specific policies and sorting. However, there are several sources of market frictions (Mortensen 2003) that contribute to wage disparities that are not captured in the AKM model, with the role of labor market structure and quality being one of them. To answer this question, I perform a variance decomposition of wages in the style of AKM to unveil how worker, firm and

market unobserved heterogeneity may impact wage inequality in Portugal. As in AKM, the largest shares of the wage variance are attributed to ability inequality (52%) and different firm pay policies (23%). However, a non-negligible 15% of the variance of real hourly wages is due to the variability in market-year effects. In other words, the fact that the same worker in the same firm may earn different wages depending on their labor market contributes almost as much to wage inequality as the variation in firms' pay policies. As for the idiosyncratic component of market structure, the heterogeneity in average quality of alternative firms is, in fact, inequality reducing, albeit negligible. Thus, alternative employers matter for individual wages and career progression, while broad differences in market pay conditions account for the important inequality margin.

1.1. Related Literature

This paper contributes to several strands of the literature. The first concerns the measurement and estimation of the value of outside options, as well as their potential impact on worker outcomes. The literature on this topic is still relatively recent. While cornerstone macroeconomic models of wage determination define the outside option of a worker as the value of non-employment (Pissarides 2000; Shimer 2010), recent evidence shows that wages may not be very sensitive to this object (Jäger et al. 2020). However, research has shown that other measures may have a direct impact on wages. Beaudry, Green, and Sand (2012) present a search and bargaining model of wage determination that considers workers' potential outside opportunities as the wages earned in their city across all industries except their own. Through a general equilibrium framework, this seminal work shows that industry composition variation across cities is closely related to average wage differences. Tschopp (2017) closely follows this strand, showing that, in Germany, a 10% increase in outside option wages generates a 7% own-wage increase. These papers rely on variation elicited by instrumental variables to isolate outside option changes from other possible confounders, such as aggregate demand. Schubert, Stansbury, and Taska (2024) focus instead on occupational mobility within a metro area to identify potential alternative wages. In this framework, a worker can leave their firm for another and remain in the same occupation, but may also switch occupations. Thus, the relevant alternative wages are a weighted average of within- and between-occupation wages. Using a unique granular dataset on occupational transitions in the US, the authors estimate an elasticity of wages to outside options of close to 0.1. This method does not, however, account for the fact that historical occupational mobility does not reflect potential, unrealized moves. More recent literature has delved into proposing alternative ways to uncover this measure that account for unobserved heterogeneity and, therefore, correct this issue. Caldwell and Danieli (2024) derive a method from a matching model with two-sided heterogeneity, concluding that a 10% increase in a worker's alternative options leads to a 1.7% increase

in wages, in Germany. Furthermore, some authors have shown that job-referral networks can impact labor market outcomes (Schmutte 2015). Following this, Caldwell and Harmon (2019) argue that coworker networks may be used to infer individual outside options. In this context, the authors find that, in Denmark, new openings at an individual's former coworkers' current firms lead to an increased probability of job transitions and also lead to wage growth. Adjacently, a recent wave of studies has focused on investigating how workers perceive their outside options, with opposite conclusions. Jäger et al. (2024) also use coworker networks as a benchmark for potential wages in the event of job dismissal, finding that workers tend to underestimate their outside option value. However, using data on actual wage offers in the U.S., Guo (2025c) reveals that workers tend to have accurate beliefs about the *average* potential wage *offered*. Using survey data from Germany, Caldwell, Haegele, and Heining (2025) reveal that the firm premia that workers expect from specific firms are highly correlated with the actual estimates of these firm premia, showcasing that workers do know about outside opportunities.

This work is also related to the monopsony literature, namely papers that study how competition for workers may influence wage markdowns. In their seminal contributions, Postel-Vinay and Robin (2002) and Cahuc, Postel-Vinay, and Robin (2006) present models of search and matching with auctions which show that firms may reduce hiring wages because they expect a steep wage schedule in the future. Through on-the-job search, workers present their current firm with alternative offers that induce competition between firms, leading to potential wage renegotiations (or separations). To insure against this, firms initially pay less than the match value to workers, generating a wage markdown. This form of monopsony power has more recently been introduced in interaction with amenities (Berger et al. 2024), market power (Jarosch, Nimczik, and Sorkin 2024) and has also been used to study wage sorting and inequality (Lamadon et al. 2024).

Third, I contribute to the peer effects literature. The reflection problem of Manski (1993) and Angrist (2014) is embedded in the estimation of outside option elasticities: if changes in the wages of firm A directly affect the wages in firms B and C, then these new wages in firms B and C may also affect the wages in firm A. The estimation of high-dimensional fixed effects models with peer effects is non-trivial. The methodology in Arcidiacono et al. (2012) is among the first to evade Manski's reflection problem in such a setting. More recently, Cornelissen, Dustmann, and Schönberg (2017) and Portugal et al. (2025) have used a non-linear least squares method to uncover the impact of co-worker ability on wages after controlling for worker and firm unobserved heterogeneity, while Baum-Snow, Gendron-Carrier, and Pavan (2024) use the same methodology to uncover local revenue and productivity spillovers at the firm level. This paper is the first to apply this empirical framework to study peer effects across workplaces.

Lastly, this paper adds to the vast literature on the determinants of wage inequality.

Abowd, Kramarz, and Margolis (1999) and Card, Heining, and Kline (2013) have pioneered the use of high-dimensional fixed effects models to study the role of worker and firm unobserved heterogeneity in affecting wage inequality. These papers show that the largest share of wage variance is explained by differences in worker effects, meaning that individual-specific characteristics (e.g., ability) drive most of the observed wage inequality. Firm-specific pay premia also have a relative importance, as does the sorting of workers to firms. While these papers have estimated the additive two-way fixed effects model for France and Germany, it has been replicated in several countries: the United States (Song et al. 2019), Brazil (Alvarez et al. 2018), Italy (Macis and Schivardi 2016; Kline, Saggio, and Sølvsten 2020), Portugal (Card et al. 2018), and others (Bonhomme et al. 2023). The standard AKM model has more recently been extended to explore other potentially important sources of heterogeneity, such as job-title heterogeneity (Torres et al. 2018), time-varying firm premia (Lachowska et al. 2023) or origin-firm heterogeneity (Di Addario et al. 2023). This paper investigates the role of another driver of inequality aside from who you are and where you work—where you could work should you leave your firm. By affecting wage and mobility decisions, the sorting of workers to different labor markets can be an important driver of wage inequality dynamics in Portugal.

1.2. Roadmap

The remainder of the paper is organized as follows. Section 2 presents the construction of labor markets, the wage equation, identification, and estimation. Section 3 describes the Portuguese matched employer–employee data. Section 4 presents the estimated markets, the wage elasticities, robustness exercises, and the evidence on current pay and future progression. Section 5 quantifies the contribution of market pay structure and sorting to wage inequality. Section 6 concludes.

2. Methodology

2.1. A stochastic block model for detecting labor markets

To detect relevant labor markets for workers, I use a degree-corrected stochastic-block model approach (Holland, Laskey, and Leinhardt 1983; Karrer and Newman 2011), a tool from the community detection literature that classifies nodes in a multigraph network into groups based on the probability of edges occurring between them. This tool can be applied in the context of labor markets, where workers or firms are assigned to "markets" based on mobility patterns across establishments and occupations. This has been used by Nimczik (2020) to identify "data-driven labor markets" based on job mobility networks in Austria. In this paper, the nodes are jobs, which are defined as an establishment ×

occupation pair, and the groups of jobs recovered by the model are interpreted as labor markets.

The reasons for defining jobs at the establishment × occupation level are threefold. First, I use establishments instead of firms because these better capture the geographical aspect of the workplace, which can be one of the most important factors driving mobility decisions. For instance, a worker may be employed by a firm that is headquartered in Lisbon and that holds several establishments across the country, with her place of work being in the Algarve. In this case, her relevant alternative choices of employment are other establishments in the Algarve region instead of Lisbon. Second, occupation is also an important factor to consider. Not only has it been used to infer workers' outside options (Schubert, Stansbury, and Taska 2024), but, when considered jointly with the establishment information, it provides a better approximation of workers' social networks. Within an establishment, workers are more likely to perform tasks in teams of workers with the same occupation, thus creating stronger ties with those co-workers. As shown in Eliason et al. (2023), the sorting of workers to establishments is highly impacted by workers' social networks, and past co-worker networks have also been used to infer outside options (Caldwell and Harmon 2019). Third, from a technical standpoint, having a stricter definition of a labor market is useful for obtaining useful identifying variation from the data.

Consider the directed job mobility network formed by the observed worker transitions. There are J jobs, indexed by $j \in \{1, \ldots, J\}$. Each observed move from job p to job q adds a directed edge from p to q; repeated moves generate multiple edges between the same pair of jobs. Let A_{pq} denote the number of observed transitions from job p to job q. Transitions in which the worker remains in the same job do not enter the network, so $A_{pp} = 0$ by construction. Each job belongs to one market $m \in \{1, \ldots, M\}$, and z_j denotes the market assignment of job j.

The model separates two sources of variation in worker flows. The first is variation in the overall number of workers entering or leaving each job. Some jobs attract many workers, while others generate many separations, irrespective of the markets to which they are connected. I capture these destination and origin propensities in γ_j^+ and γ_j^-, respectively. The second source of variation is the pattern of flows between markets. Let $\mathbf{B}$ be an $M \times M$ matrix describing the strength of mobility flows between labor markets. Each entry B_{uv} measures the intensity of worker transitions from jobs in market u to jobs in market v, after accounting for the fact that some jobs generate more separations and others hire more. A high value of B_{uv} means that workers move from market u to v more frequently than would be expected solely from the overall number of workers leaving jobs in u or entering jobs in v.

Conditional on the market assignments and model parameters, I assume that the

transition counts are independent Poisson random variables:

ASSUMPTION 1. $A_{pq} \mid \mathbf{B}, \mathbf{z}, \boldsymbol{\gamma}^{+}, \boldsymbol{\gamma}^{-} \sim \text{Poisson}\left(\gamma_p^{-}\gamma_q^{+}B_{z_p,z_q}\right), \qquad p \neq q.$

Hence, the expected number of transitions from job p to job q is

$$\mathbb{E}\left[A_{pq} \mid \mathbf{B}, \mathbf{z}, \boldsymbol{\gamma}^{+}, \boldsymbol{\gamma}^{-}\right] = \gamma_p^{-}\gamma_q^{+}B_{z_p,z_q}. \tag{2}$$

In logs:

$$\log \mathbb{E}[A_{pq}] = \log \gamma_p^{-} + \log \gamma_q^{+} + \log B_{z_p,z_q}. \tag{3}$$

where z_p and z_q represent the markets of job p and job q, respectively. This resembles a gravity or trade model, in which the flow count between two jobs is given by an origin effect governing how strongly it generates worker outflows, a destination effect capturing how strongly the destination attracts workers, and a dyad fixed effect that captures the strength of the connection between markets.

The degree correction is important in this application because jobs differ greatly in size and turnover. Without γ_p^{-} and γ_q^{+}, the model could assign large and small jobs to different markets simply because they generate different numbers of transitions. With the correction, jobs are grouped according to their block-level mobility patterns after accounting for these differences in total inflows and outflows. Thus, jobs assigned to the same market share a similar pattern of connections to other markets, even if they do not have the same number of accessions or separations.

To separate the scale of the degree parameters from the block-to-block intensities, I impose the normalizations

$$\sum_{j:z_j=m} \gamma_j^{-} = 1 \qquad \text{and} \qquad \sum_{j:z_j=m} \gamma_j^{+} = 1 \qquad \text{for every market } m. \tag{4}$$

The likelihood associated with the model is

$$\mathcal{L}\left(\mathbf{B}, \mathbf{z}, \boldsymbol{\gamma}^{+}, \boldsymbol{\gamma}^{-}; \mathbf{A}\right) = \prod_{p \neq q} \frac{\left(\gamma_p^{-}\gamma_q^{+}B_{z_p,z_q}\right)^{A_{pq}}}{A_{pq}!} \exp\left(-\gamma_p^{-}\gamma_q^{+}B_{z_p,z_q}\right). \tag{5}$$

In the community-detection literature, this representation is called the canonical formulation. Conditional on $\boldsymbol{\gamma}^{-}$, $\boldsymbol{\gamma}^{+}$, and $\mathbf{B}$, both the number of transitions associated with each job and the number of transitions between each pair of markets remain random. This is

analogous to a Poisson count model in which the covariates determine the expected number of events, but the observed count need not equal its conditional mean. The canonical formulation is useful because it makes the economic content of the model transparent: observed mobility depends both on job-specific inflow and outflow propensities and on the strength of the connection between the origin and destination markets. It is a model of realized worker transitions and does not require job offers themselves to be observed or to arrive according to a Poisson process.

I estimate a version of this model that imposes the same restrictions *exactly* rather than in expectation. In practice, it conditions on the *observed* number of transitions entering and leaving each job and, for any candidate partition, on the *observed* number of transitions between each pair of markets. The remaining randomness concerns how those transitions are allocated across individual origin–destination job pairs, subject to these totals. This is called the microcanonical formulation. In other words, the canonical formulation uses $\boldsymbol{\gamma}^{-}$, $\boldsymbol{\gamma}^{+}$, and $\mathbf{B}$ to describe expected network totals, whereas the microcanonical formulation replaces them with their realized counterparts: the job degree sequence and the matrix of block-to-block transition counts. The two formulations preserve the same economic distinction between job-specific mobility and market-level connections, but differ in which quantities are treated as random.

In estimation, I use the microcanonical representation of the degree-corrected block model proposed by Peixoto (2014). Appendix B presents the likelihood and its relationship to equation (5). The microcanonical formulation is useful because it avoids the estimation of thousands of nuisance parameters, since we have to estimate, for each job, inflow and outflow propensities and, for each dyad, one fixed effect. Furthermore, estimation of (5) favors more detailed partitions, which is not necessarily desirable. The description-length estimator detailed in the appendix is useful because it forces the estimator to choose which jobs belong together and how many markets there should be while balancing fit and complexity.

2.2. Defining relevant alternative employers

After classifying jobs into markets, it is then possible to define workers' relevant alternative options, and thus understand the impact of the job network structure on wages.

Starting from the markets defined in the previous section, I delimit the job outside options for a given worker to the establishments in those markets that have at least one worker with the same occupation in a given year. To put it differently, consider worker i, working in establishment e, with occupation o. This establishment × occupation pair belongs to market m. So, the worker's job outside options are jobs in all the establishments of market m that have at least one worker with occupation o. This assumption implies that workers only receive offers from those establishments and, therefore, do not move

across markets or occupations. While this is not realistic in the data, it helps pin down the relevant network structure in a parsimonious way.

2.3. A peer effects perspective of wage interdependence

After defining the set of firms that constitute a worker's relevant employment alternatives, I investigate whether the pay policies of these firms are related to the wage she receives in her current job. To do so, I use a peer effects framework and extend the usual two-way fixed effects equation of Abowd, Kramarz, and Margolis (1999). The main idea is that a worker's wage may depend not only on who she is and where she currently works, but also on the firms she could potentially work for. My main specification is

$$w_{ifmt} = \mathbf{x}'_{ifmt}\boldsymbol{\beta} + \alpha_i + \theta_{mt} + \zeta_f + \gamma\overline{\zeta}_{-fmt} + \varepsilon_{ifmt}, \tag{6}$$

where w_{ifmt} is the wage of worker i in firm f, market m, and year t; $\mathbf{x}_{ifmt}$ is a vector of observable characteristics; α_i is a time-invariant worker wage component; ζ_f is the wage premium associated with the worker's current firm; and θ_{mt} is a market-year fixed effect. The final object, $\overline{\zeta}_{-fmt}$, is the average wage premium of all other firms in the worker's market, excluding firm f. In the main specification, this average is weighted by the employment share of each firm in the market.

The leave-out average, $\overline{\zeta}_{-fmt}$, provides a summary of the pay quality of the worker's alternative employers. A market in which high-premium firms account for a large share of employment will, in principle, provide more attractive alternatives than one dominated by low-premium firms. The measure should not, however, be read as the value of a particular job offer or as a *complete* measure of the worker's outside option. Workers' outside options may depend not just on what other firms may pay, but also what they offer in terms of amenities and on the costs of switching jobs. This formulation also does not take into account the fact that some firms may not be hiring. Thus, the measure of outside options considered in this paper—and the elasticity γ—represent how changes in potential alternative employment opportunities impact wages.

There are several reasons that explain why wage interdependence exists, and this model is agnostic about which one dominates. Models of industrial composition, such as Beaudry, Green, and Sand (2012) and Tschopp (2017), show that the wages of workers in a given industry and city are positively correlated with the wages of all other industries. Hence, these models would predict a positive elasticity of wages to alternative firms' pay policies, i.e., $\gamma > 0$. Because the SBM can cluster firms from different industries into the same market, this mechanism could very well be captured by γ in my model. The search and matching models with wage renegotiation and bargaining, as pioneered by

Postel-Vinay and Robin (2002) and Cahuc, Postel-Vinay, and Robin (2006), show different possible predictions. In these models, because workers can search on the job and obtain more favorable offers, firms can initially pay lower wages to back-load future possible increases from renegotiation, provided that they have some bargaining power. This "quit/renegotiation" term would predict that wages are decreasing in other firms' pay policies, i.e., $\gamma < 0$. This can also be seen as a compensating differential (Lamadon et al. 2024), as workers are (negatively) compensated for future increases in their position. However, bargaining models also show that workers receive a share of their outside option, be it measured as the lost value of unemployment (Berger et al. 2024) or as their next best offer Caldwell and Danieli (2024). In this case, positive changes in the quality of other firms' policies through the structure of the market impact workers' outside options, driving up wages. These two mechanisms counteract each other and the net effect on wages is ambiguous. The leave-out average of firm effects may also reflect changes in the probability of receiving better outside offers due to improved information about the market. By having better knowledge about potential outside offers, workers ameliorate their search and may, therefore, have a higher probability of receiving a good offer (Jäger et al. 2024). This mechanism, recently explored in Caldwell and Harmon (2019), would predict that $\gamma > 0$. Lastly, human capital accumulation and learning about worker or match quality (Gualdani et al. 2026) may also influence the sign of the coefficient. All in all, it is unclear what direction γ should take.

Equation (6) separates two dimensions of market pay structure. The market-year effect, θ_{mt}, captures wage determinants that are common to all workers in the same market and year. These may include local labor demand and supply conditions, occupation-specific trends, institutions, and other shocks that affect the market as a whole. In contrast, $\overline{\zeta}_{-fmt}$ captures differences in the composition of firms outside the worker's current employer. Two workers may belong to the same market in the same year but face different leave-out averages because they work at different firms. For example, a worker who is already employed by the largest high-paying firm in the market excludes that firm from her set of alternatives. Another worker in a smaller low-paying firm includes it, and may therefore face a considerably better set of alternative employers. The two workers share the same general market conditions, but their position within the market is different.

This is where the peer effects interpretation is relevant. Firms are connected because they belong to the same labor market and compete for similar workers. The model allows the wage paid by one firm to be related to the wage-setting characteristics of the other firms in that market. The parameter γ, therefore, summarizes this cross-firm wage interdependence. Conditional on the worker, current firm, and common market-year component, it asks whether wages differ when the pay quality and employment shares of the remaining firms in the market differ.

For comparison, I also estimate a restricted version of the model that omits the market-year fixed effects:

$$w_{ifmt} = \mathbf{x}'_{ifmt}\boldsymbol{\beta} + \alpha_i + \zeta_f + \gamma\bar{\zeta}_{-fmt} + \varepsilon_{ifmt}. \tag{7}$$

In this specification, γ combines the relative pay quality of alternative firms with market-year conditions that are correlated with that quality. For instance, the expansion of high-premium firms may coincide with a general improvement in labor demand that raises wages throughout the market and, without θ_{mt}, the model cannot distinguish the two sources of variation. I treat equation (7) as a benchmark, and compare it with equation (6) to show how the estimated relationship changes once common market-year wage determinants are absorbed.

2.3.1. Identification

There are two parts to the identification discussion. The first concerns the worker, firm and market-year fixed effects, α_i, ζ_f and θ_{mt}. The second concerns γ, which relates workers' wages to the pay quality of their alternative employers. Separating these objects requires both links between workers, firms and markets and variation in the composition of the firms that workers could move to.

Consider the following assumptions:

ASSUMPTION 2. ***Exogenous mobility***.

Let $f(i,t)$ and $m(i,t)$ denote the firm and market assignment of worker i at time t. Let $\omega_{m(i,t)} = \{\omega_1, \ldots, \omega_F\}$ be the set of weights given to each of the F firms in market $m(i,t)$. Lastly, let $\Omega_i = \{\alpha_i, f(i,t), m(i,t), X_{it}, \omega_{m(i,t)}\}_{t=0}^{T_i}$. Then, for the set of error terms $\varepsilon_i = \{\varepsilon_{i,f(i,1),m(i,1),1}, \ldots, \varepsilon_{i,f(i,T_i),m(i,T_i),T_i}\}$

$$\mathbb{E}\left[\varepsilon_i \mid \Omega_i\right] = 0 \tag{8}$$

This is a standard strict exogeneity assumption which, in the context of AKM models, is also denominated as an exogenous mobility assumption. It states that the wage errors, ε_i, should have an expected value of zero conditional on worker type, firm and market assignment, covariates and other firms' weights (used in the leave-out average firm effect). This is an important assumption that restricts mobility decisions to be uncorrelated with firm- or market-match effects. At the same time, this does not rule out any effects of worker, firm or market on mobility decisions. It may still be possible that high α workers move more frequently than low α ones, or that they move to firms with higher premia than their counterparts.

ASSUMPTION 3. ***Additivity***.

Conditional on the observable characteristics and the leave-out average of firm effects, the worker, firm and market-year wage components enter additively. There are no additional systematic interactions between these components in the conditional mean of wages.

This assumption rules out match-specific effects in which any interactions between the fixed effects are relevant. In a potential outcomes framework, it is equivalent to saying that the potential outcomes from working at firm f and market m equal the sum of the potential outcomes of firm f and market m.

Given the stated assumptions, the identification argument for α, θ and ζ is similar to that of Abowd, Creecy, and Kramarz (2002) but with an extra set of fixed effects.

DEFINITION 1. *Consider three-way tuples of worker, firm and market-time (w, f, mt). Two tuples are* ***nearly identical*** *if they are similar in all but one element of the tuple.*

I prune the data to the set of three-way tuples (worker, firm, market-time) that are nearly identical. For example, the tuples (worker 1, firm 1, market-time 1) and (worker 2, firm 1, market-time 1) are nearly identical. By classifying all the tuples iteratively, it is possible to create sets of tuples that are connected by this definition. I follow Weeks and Williams (1964) and Stata's *group3hdfe* command (Guimaraes 2016) to obtain the largest connected set of tuples that grants identification of the fixed effects.

Workers who remain at the same firm and in the same market can also contribute to the estimation. Their wage changes help distinguish changes in the common market-year component from changes associated with the composition of their alternative employers. However, these workers do not, by themselves, separate their permanent worker component from their employer's time-invariant premium. The links provided by movers remain relevant for that comparison.

Next, we have the identification of γ. This is obtained through changes in the job network composition, i.e., variation in the peer groups' composition. To see where this variation comes from, let n_{jmt} denote employment at firm j in market m and year t. The employment-weighted average can be written as

$$\bar{\zeta}_{-fmt} = \frac{\sum_{j \in \mathcal{F}_{mt},\, j \neq f} n_{jmt} \zeta_j}{\sum_{j \in \mathcal{F}_{mt},\, j \neq f} n_{jmt}}, \tag{9}$$

where $\mathcal{F}_{mt}$ is the set of firms in the relevant market-year. Throughout this discussion, the market incorporates the same-occupation restriction used to define relevant alternative employers. The current firm is excluded from both the numerator and the denominator, including any of its establishments in that market.

Consider a market with three firms, A, B and C, where A has the highest premium. If A expands relative to C, the leave-out average facing workers at B increases. Workers at A,

however, exclude A from their alternatives, so A's own expansion does not directly change their leave-out average. Thus, the same change in market structure can affect workers' measured alternatives differently depending on their current employer. Firm entry into or exit from the market, changes in rivals' relative employment, and differences in the alternatives faced by the same firm across markets can provide similar variation. Because the firm effects are time-invariant, these are changes in exposure to firms with different premia, rather than yearly changes in the premia themselves.

Consider a situation in which all firms have equal weights. Let F_{mt} denote the number of firms, rather than employment. Then

$$\bar{\zeta}_{-fmt} = \underbrace{\frac{\sum_{j \in \mathcal{F}_{mt}} \zeta_j}{F_{mt} - 1}}_{a_{mt}} - \frac{\zeta_f}{F_{mt} - 1}. \tag{10}$$

The market-year effects absorb wage changes that are common to workers in the same market and year. Hence, they absorb a_{mt}. The firm effects also account for permanent wage differences between employers. Identifying variation, therefore, is generated solely through $\frac{1}{F_{mt}-1}$. If $F_{mt} = F^*$, the firm effect captures $\frac{1}{F^*-1}$ and no variation remains.

However, unequal firm sizes alone do not identify γ. If every firm always faces the same alternatives in the same proportions and operates in only one market, its leave-out average is constant and cannot be distinguished from its permanent wage component. Let $S_{mt} = \sum_{j \in \mathcal{F}_{mt}} \omega_{jmt} \zeta_j$. The weighted index satisfies

$$\begin{aligned} \bar{\zeta}_{-fmt} &= \frac{S_{mt} - \omega_{fmt}\zeta_f}{1 - \omega_{fmt}} \\ &= S_{mt} + \frac{\omega_{fmt}}{1 - \omega_{fmt}} \left(S_{mt} - \zeta_f \right). \end{aligned} \tag{11}$$

The market-year effect absorbs S_{mt}. In a static economy with single-market firms, the firm fixed effects absorb the remaining variation. Thus, the model needs exposure differences that remain informative after these wage components are accounted for. In other words, the identifying variation is sourced from 1) firms that are present in multiple markets, and 2) changes in market structure over time.

Without market-year effects, equation (7) also uses exposure variation associated with common market conditions. For example, high-premium firms may expand during a local demand boom that raises wages throughout the market. Equation (6) absorbs the common wage component and asks whether wages are related to the remaining differences in alternative-employer pay quality. These specifications therefore describe

different relationships.

The available variation does not, on its own, establish a causal effect of outside options. For example, a high-premium firm's expansion may coincide with a fall in demand for another firm's product. Workers at the latter firm would then face a higher leave-out average and lower wages, even without a response to better employment alternatives. Market-year effects absorb shocks that affect wages equally across the market, but do not remove firm-specific shocks or differences in firms' responses to a common shock. These concerns motivate the robustness exercises in section 4.4.

2.3.2. Estimation

I estimate equations (7) and (6) by nonlinear least squares, following the panel spillover approach of Arcidiacono et al. (2012) and its implementation with high-dimensional fixed effects in Portugal et al. (2025). The firm premia and the response to alternative employers' pay quality are estimated jointly: the same firm effects enter both the current employer's wage component and the leave-out average. Conditional on a set of firm premia, I estimate the observable variables' coefficients, worker and market-year effects, and γ through a linear regression. I then update the firm premia conditional on those estimates and repeat these steps. Appendix C presents the matrix representation and estimation equations.

2.3.3. Bias correction

Estimation with many fixed effects can produce finite-sample bias when the information available to estimate each effect is limited (Neyman and Scott 1948). To reduce potential estimation bias, I apply a split-panel jackknife correction based on Dhaene and Jochmans (2015). The procedure compares estimates obtained from the full panel with estimates obtained from two shorter panels.

For each worker, I order the observed work history chronologically and divide it into two consecutive halves. I re-estimate the model separately in each half, applying the same support and connected-set restrictions used for the full sample. The firm premia and leave-out averages are re-estimated within each half-panel. Each half must retain enough observations and worker–firm–market links to estimate the model.

Let $\vartheta = (\beta', \gamma)'$ collect the common slope parameters. Denote their full-panel estimate by $\hat{\vartheta}_{\text{full}}$ and the two half-panel estimates by $\hat{\vartheta}_1$ and $\hat{\vartheta}_2$. The bias-corrected estimate is

$$\hat{\vartheta}_{\text{BC}} = 2\hat{\vartheta}_{\text{full}} - \frac{1}{2}\left(\hat{\vartheta}_1 + \hat{\vartheta}_2\right). \tag{12}$$

The rationale is that, when the leading bias is inversely proportional to panel length and comparable across the two halves, shortening the panel approximately doubles that

component. Equation (12) is designed to remove it. Higher-order finite-sample bias can remain, so I refer to these estimates as bias-corrected. The correction is applied separately to each wage specification, and I report the full-panel and bias-corrected estimates together.

3. Data

I use Quadros de Pessoal (QP, Lists of Personnel), a unique matched employer-employee administrative dataset sourced from the Ministry of Employment and Social Security. This is a mandatory survey filled in by every establishment having at least one wage-earner, containing information on virtually the universe of private Portuguese firms and their employees. The survey is responded to in October of each year (March prior to 1993). The years available are 1986-2021, with gaps for 1990 and 2001, although the estimation considers only the period 2010-2021. Being a compulsory annual survey, it ensures that problems such as panel attrition are minimized. Furthermore, measurement error, missing values and misclassification issues are also rare due to its handling by the Portuguese authorities. Notwithstanding, this is a survey of employer firms: it does not include information on civil servants, self-employed or the unemployed. Its coverage of the agricultural and fisheries industries is also lacking due to the high prevalence of self-employed and informal work.

Each firm entering the database is assigned a unique identifying number, allowing researchers to track the same firm over time. Each worker is also identified with a unique number, enabling the matching of workers to firms. It is, therefore, possible to build a worker's employment history as long as they remain employed by firms targeted by the survey.

The dataset contains information about the firm's industry, sales, employment, location, ownership, and legal status. It also provides information on establishments, which are the smallest units belonging to a firm that can employ workers, and can be linked to employees with a unique identifier. This is important to define relevant labor markets given the potentially geographical distance between a firm's headquarters and an employee's workplace. For example, a firm headquartered in Lisbon can have several establishments across the country. Failure to account for this would lead us to conclude that some employees' relevant labor market would be based in Lisbon, whereas they may be in other parts of the country. Data on workers includes monthly wages (including base, overtime, and supplements), hours worked (regular and overtime), demographics (age, gender, education, nationality), occupation (ISCO 4-digit), and starting date with the firm.

I employ some restrictions on the dataset to identify and estimate the model of interest. First, the final sample was restricted to full-time workers aged 18-64 who earn at least 80% of the national minimum wage. This threshold is motivated by the fact that in the

Portuguese labor market, apprenticeships may collect 80 percent of the minimum wage. Second, only workers in the services and manufacturing sectors were included, implying the exclusion of those working in agriculture, fisheries, energy, and extraction sectors. Third, for estimation purposes, only peer groups (defined in the next section) with more than two firms and workers are kept. Lastly, I exclude workers and firms not belonging to the largest three-way connected set between worker, firm and peer group. This latest exclusion restriction is needed for the identification of firm and peer fixed effects, which are only comparable between firms and peer groups linked by mobility (see Section 2.3.1 for details). The last two steps are done in an iterative manner to ensure all the exclusion restrictions are met in the final dataset.

Overall, the final sample includes 16,064,983 observations, corresponding to 3,249,865 workers and 203,304 firms. Based on the aforementioned restrictions, this sample's observations correspond to 68% of the original dataset.

4. Results

4.1. Markets summary statistics

The estimation of the job mobility network in Section 2.1 yields, for each establishment-occupation pair (i.e., job), a market assignment. Table 1 highlights the summary statistics of this process, estimating a total of 688 markets across the country, with each market having, on average, 615 firms. Firms may be present in more than one labor market, either because they are multi-establishment firms or because they employ workers across several occupation groups. This is reflected in the summary statistics: firms participate, on average, in 15.3 markets and they have an average of 19.3 workers in each of these markets.

Lastly, 37% of firm-to-firm transitions occur within the same market, a percentage that goes up to almost 49% if, instead, we consider moves across establishments. According to the definition of a job, workers may also move to another job while remaining in the same firm or establishment. Thus, considering all the job-to-job flows, the share within a market sits at 68%. It is important to note that the SBM does not mechanically maximize the share of movers within markets. It groups jobs according to their mobility patterns, accounting for differences in the number of workers entering and leaving each job. Thus, the observed concentration of job-to-job flows within the estimated blocks supports their interpretation as labor markets, although moves across blocks remain important.

4.2. Firm and market effects

The estimated firm and market-year effects, $\hat{\zeta}$ and $\hat{\theta}$, capture persistent differences in employers' pay premia and wage determinants common to workers in the same market

TABLE 1. Markets summary statistics

Statistic	Value
Number of markets	688
Average firms per market	614.8
Average firm size per market	19.3
Average markets per firm	15.3
Firm-firm flows within market	37.0%
Establishment-establishment flows within market	48.7%
Job-to-job flows within market	68%

Notes: All statistics refer to the year of 2021 with the exception of the last two rows, which are obtained from the whole sample.

and year, respectively. Given that these effects are identified up to a constant, the literature usually normalizes them to enable a potential interpretation. The firm effects are normalized such that the average of the Accommodation and Food Services industry – a sector with low firm premia – is zero, while the market-year effects are normalized such that the average of the 2010 effects is zero. Appendix D presents the distributions of firm and market effects and shows how the market effects change with occupation, sector, region and time. I also show that high-wage workers tend to work for higher-premium firms, while sorting across markets is weaker but changes over time. The contribution of sorting to wage inequality is discussed in Section 5.

Leave-out average firm effects The other object of interest from equation (6) is the leave-out average of firm effects, $\overline{\zeta}_{-fmt}$. This measures the average pay premium of alternative employers in the worker's market and occupation, excluding her current firm and weighting each alternative by its employment. The measure therefore summarizes the pay quality of potential alternative employers, without requiring that every worker receives or accepts an offer from each of them.

Table 2 presents the mean and median of the estimated $\overline{\hat{\zeta}}_{-fmt}$ for the full sample and selected groups of workers. The full-sample mean is 0.18 log points and the median is 0.16 log points, relative to the normalization of firm effects in Accommodation and Food Services. These are levels of alternative-employer pay premia, rather than differences relative to the current employer.

When looking at the differences across wage groups, higher-paid workers tend to face higher-premium alternative employers. At the bottom of the wage distribution, the average alternative premium is 0.11 log points, compared with 0.30 log points for workers in the top decile. This is consistent with higher-wage workers having access to better-paying employers, but does not show that these alternatives pay more than their current firm. A

TABLE 2. Summary statistics of leave-out average firm effects, by group

	Mean	Median
Full sample	0.1771	0.1642
Wage quantiles		
Below P25	0.1088	0.1029
P25 to P50	0.1482	0.1443
P50 to P90	0.2072	0.2004
Above P90	0.2997	0.2798
Labor market concentration		
Low concentration	0.1405	0.1328
High concentration	0.2138	0.2014

Notes: Mean and median of the leave-out average of estimated firm effects, in log points. Alternative firms are in the same SBM market and occupation; the current firm and all its establishments are excluded. The average is weighted by alternative firms' employment. Firm effects are normalized such that their average in Accommodation and Food Services is zero. Wage quantiles are defined within year. Low concentration = market-year HHI below the yearly median.

worker may already be employed by a high-premium firm and still face alternatives that pay less than her current employer.

The leave-out average is also higher in more concentrated markets: 0.21 log points, compared with 0.14 log points in less concentrated markets. This does not mean that in concentrated markets, workers have more outside options, as we are comparing an average term. Thus, while concentrated labor markets have fewer alternative employers, their pay premia are, on average, higher. Appendix D presents the detailed summary statistics, including the education, gender and collective bargaining splits.

Does the measure predict destination firm premia? To illustrate how predictive $\bar{\hat{\zeta}}_{-fmt}$ is of job movers' new firm premia, I regress, for those who switch firms within the same market, the destination firm's premium on the estimated leave-out average in the worker's origin firm and market in the previous year. Figure 1 illustrates this relationship in a scatter plot, showing a regression slope of 0.78 and a correlation coefficient of 0.52. Workers facing higher-premium alternative employers tend to move to firms with higher premia. The relationship is also positive, yet weaker, when I include workers who move across markets (see Figure 12 in the appendix).

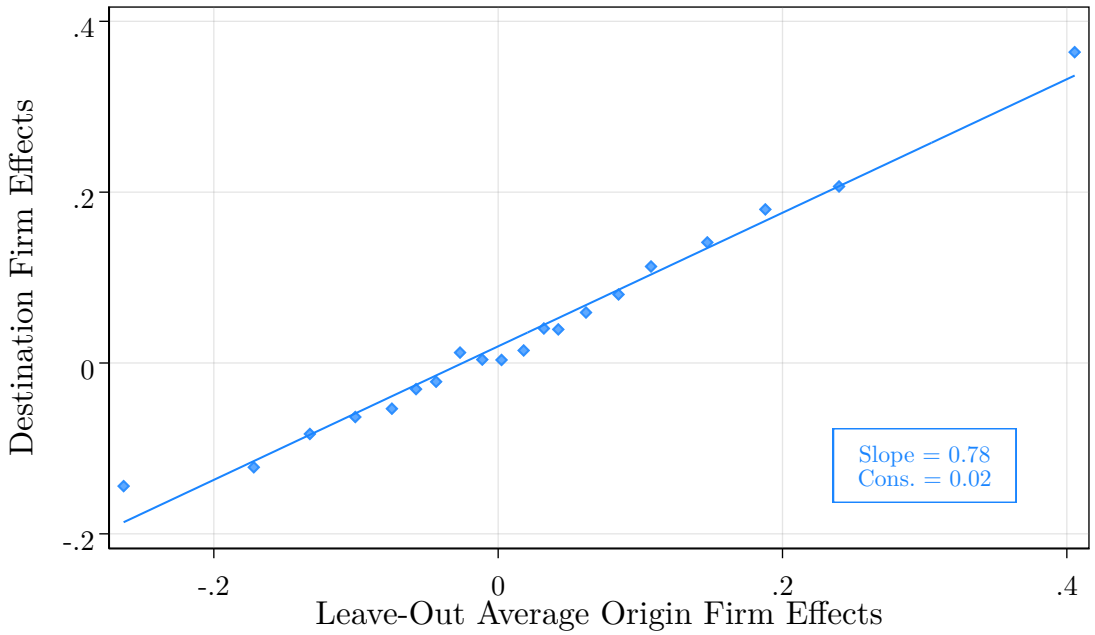


FIGURE 1. Destination firm effects and the leave-out weighted average

Notes: Workers who switch firms within the same market. The figure relates the destination firm's estimated premium to the employment-weighted leave-out average in the worker's origin firm and market in the previous year.

This positive relationship supports the leave-out average as a summary of alternative

employers' pay quality. The comparison concerns observed movers, who may reject lower-paying offers or search directly for particular firms, so destination firms need not be a random draw from all available alternatives. Nonetheless, employment weighting has a grounding in models such as those in Beaudry, Green, and Sand (2012) and Tschopp (2017), in which the probability of meeting a particular employer can be related to its employment share in the market.

I next investigate how these alternative employment opportunities may shape wages.

4.3. The impact of other employers' pay policies on wages

How do changes in the quality of other firms' pay policies affect a worker's wage?

Table 3 presents the results of estimating the models in equations (7) and (6). Annex Table 12 presents all the estimated coefficients. The first two columns show the results without the market-year fixed effects. The table presents bias-adjusted coefficients in columns 2 and 4 as discussed in section 2.3.3, which are the main coefficients of interest. It shows that, after bias correction, a 10% increase in the quality of alternative firms is associated with an increase in wages, on average, of about 2.8%. This elasticity lies slightly above the estimates of 1% in Schubert, Stansbury, and Taska (2024) and 1.7% in Caldwell and Danieli (2024), but below the larger effect of 7% found in Tschopp (2017). However, the inclusion of the market-year fixed effect flips the sign, with the elasticity dropping to -0.1. These effects are statistically significant at any reasonable significance level, with standard errors clustered at the firm-year level.[1] Another intuitive way to think about the magnitude of these coefficients is to interpret them based on the variation in the quality of firm effects within markets. To do so, I calculate the standard deviation of the leave-out estimated firm effects for each market-year pair, and then average them across the sample. I then multiply this standard deviation by the respective coefficient to obtain an interpretation of γ in terms of this variation. A one-standard-deviation increase in the quality of outside options adjusts wages by 3.62% in the case without market-year effects and -1.25% in the full model, after bias adjustments.

Why does accounting for common market conditions change the relationship so much? Consider a local demand boom in which high-premium firms expand and wages rise throughout the market. Workers would face better-paying alternatives at the same time as their own wages increase. The specification without market-year effects can capture this positive comovement, alongside any wage response to the changing alternatives. The market-year effect absorbs the wage component shared by workers in that market and year. The second specification therefore asks a narrower question: how do wages relate to differences in alternative-employer pay quality after common market conditions

[1] Clustering the standard errors at the worker, firm, or market-year level does not change the statistical significance of the result.

TABLE 3. Elasticity of wages to other firms' pay premia

	(1)	(2)	(3)	(4)
$\hat{\gamma}$	0.2117***	0.2784***	-0.0731***	-0.1002***
	(0.0038)	(0.0073)	(0.0045)	(0.0063)
Worker FE	✓	✓	✓	✓
Firm FE	✓	✓	✓	✓
Market-Year FE	–	–	✓	✓
Split-Sample?	–	✓	–	✓
Weighted?	✓	✓	✓	✓
Observations	16,064,983	16,064,983	16,064,983	16,064,983

Note: Controls for age squared, education and a quadratic function of tenure. Standard errors clustered at firm × year level. Split-sample refers to incidental parameter bias adjustment as in Dhaene and Jochmans (2015). See Section 2.3.3 for details. Weighted refers to current employment weights used in the computation of the leave-out average of firm effects.
*** Denotes statistical significance at the 0.01 level.

are accounted for? These differences arise partly because workers exclude their own employer from the measure. For example, if a high-premium firm expands while other firms' employment is unchanged, its larger employment share raises the leave-out average facing workers at other firms. Its own workers exclude it from their alternatives, so its expansion does not directly change their index. Firm entry, exit and changes in relative employment can generate similar differences. Since the estimated firm premia are time invariant, these movements reflect changing exposure to employers with different premia, rather than directly observed yearly changes in their pay policies.

4.4. Robustness

Before considering possible explanations, I examine whether the sign reversal depends on the employment weights or the definition of alternative employers. The employment weights used in the main estimations may be endogenous. Market-level employment responds to firm-specific shocks to demand, productivity, recruiting effort, and other variables. Our current specification, by not allowing for a firm-market-year effect, does not control for these shocks, which may be correlated with worker-level wages and employment. Thus, the coefficients seen in Table 3 may fail a credibility test.

To investigate this, we estimate equation 6 using three additional sets of weights. First, I use the previous year's employment as a weight. This insures our estimates against the endogeneity that may arise from current period shocks affecting current employment, as well as removing any concerns regarding contemporaneous reverse causality. Second, inspired by shift-share instruments, I use the employment growth in all other markets

TABLE 4. Elasticity of wages to other firms' pay premia – Alternative weights

Employment weights	Without market-year FE	With market-year FE	Observations
Current	0.2784***	-0.1002***	16,064,983
	(0.0073)	(0.0063)	
Lagged	0.2333***	-0.1000***	13,919,785
	(0.0060)	(0.0074)	
Leave-out market predicted	0.1818***	-0.0638***	13,359,886
	(0.0056)	(0.0072)	
Sector employment growth predicted	0.2264***	-0.0899***	13,830,130
	(0.0060)	(0.0073)	

Note: Current and lagged weights use employment in the current and previous year, respectively. Leave-out market predicted weights update lagged employment using the firm's employment growth in its other markets. Sector employment growth predicted weights instead update lagged employment using national employment growth in the firm's main sector, excluding the markets in which it operates. Split-sample results. Standard errors are clustered at the firm × year level.
*** Denotes statistical significance at the 0.01 level.

of the firm as a predictor for employment growth in a given market, and apply that predicted growth to the previous period employment levels. Thus, we obtain a current period predicted employment weight, which is exempt from market-specific interactions that may simultaneously affect wages and employment shares. Third, I employ the same predicted employment strategy, but considering national employment growth in the firm's main sector of activity, excluding the growth in its own markets, to predict current employment. This way, current employment depends only on the lagged employment of the firm and the national trends in its sector, which removes potential endogeneity stemming from firm-specific demand shocks.

The first row of Table 4 reproduces the original results, while the remaining rows show the alternative weighting schemes. The elasticities remain similar to the ones in Table 3, lending qualitative strength to the results. Without market-year fixed effects, we see a positive effect of outside options on wages; with market-year fixed effects, the sign flips, as was the case in Table 3. The results are, therefore, robust to the specific forms of endogeneity tackled in this exercise.

The estimation of the elasticity of wages to the quality of alternative firms hinges on the assumption that the average of other firm effects is computed with the correct firms, i.e., that the peer group is well defined. The peer group is characterized by the intersection between the markets obtained from the clustering algorithm of the SBM in Section 2.1 and occupation, with the assumption being that workers tend to search for jobs in similar occupations. Table 13 checks the robustness of the results to other group definitions.

The conditional relationship remains negative using either the SBM blocks without the additional occupation restriction or occupation-by-NUTS-II markets. The latter groups workers in the same occupation and broad region even if their jobs belong to different mobility blocks. The estimates are −0.269 and −0.126. Given that the markets are larger in these definitions, it is also useful to put these numbers in terms of standard deviations. A one-standard-deviation increase in the leave-out average terms is associated with a decrease in wages of 2% in the first case and 1.3% in the second. Thus, the negative sign survives these definitions, although its magnitude varies, especially when we consider the broader SBM-made groups.

4.5. Heterogeneity

Appendix E examines heterogeneity by education, occupation, industry, and labor market concentration. While the negative wage elasticity is not driven exclusively by a single group, the effects of alternative pay quality are higher in low-skilled occupations and for low-educated workers, which can be a consequence of these workers' lower bargaining power. Nevertheless, the negative effect is present across most occupation groups and major industries, as well as in both more and less concentrated labor markets.

4.6. Understanding the negative wage elasticity

The negative conditional wage relationship raises a question: why would workers earn less when better-paying employers become more important among their alternatives? The answer may lie in the distinction between the wage a job pays today and the opportunities it creates for tomorrow. A worker who expects employment to lead to better offers, stronger bargaining positions, or more valuable skills may accept lower current pay. The evidence points toward this broad dynamic compensation mechanism: the negative relationship is strongest near hiring, occurs even among experienced new hires, and accompanies higher subsequent wage growth among workers who remain with their employer.

4.6.1. Firm shocks and workplace composition

A first explanation concerns the shocks behind changes in alternative firms' employment shares. Consider two manufacturers competing for the same customers. If the higher-paying manufacturer wins a large order and expands, it becomes more prominent among the other manufacturer's employment alternatives. At the same time, the loss of orders may put downward pressure on wages at the other firm. The resulting negative relationship could partly reflect a shift in product demand. Market-year effects absorb common wage conditions, but they cannot remove every shock that affects employers differently. This scenario would constitute a violation of our exogeneity assumption, as we would be

associating the firm's negative demand shock with an improvement in the leave-out average of firm effects term.

In the robustness section, we already saw that the results are robust to different weighting schemes, as lagged employment or predicted employment weights still yield negative elasticities (with similar magnitudes). These exercises do not eliminate persistent or correlated firm shocks, however. I therefore examine whether the relationship is concentrated among firms that are more likely to compete for the same customers.

Let $a_{j|-f,mt}$ be firm j's employment share among all other employers in market m and year t, excluding the current firm f. If I_{jt} denotes firm j's industry, the two contributions are

$$z^{S}_{-fmt} = \sum_{j \neq f} a_{j|-f,mt}\, \zeta_j \mathbf{1}\{I_{jt} = I_{ft}\},$$
$$z^{D}_{-fmt} = \sum_{j \neq f} a_{j|-f,mt}\, \zeta_j \mathbf{1}\{I_{jt} \neq I_{ft}\}.$$

I propose estimating

$$w_{ifmt} = \mathbf{x}'_{it}\boldsymbol{\beta} + \alpha_i + \zeta_f + \theta_{mt} + \gamma_S z^{S}_{-fmt} + \gamma_D z^{D}_{-fmt} + \varepsilon_{ifmt}, \tag{13}$$

where w_{ifmt} is log wages, $\mathbf{x}_{it}$ contains the main wage controls, and α_i, ζ_f, and θ_{mt} are worker, firm, and market-year effects.

Table 5 shows that the elasticity is stronger for firms within the same labor market but which do not belong to the same industry. However, a strong, negative elasticity of -0.08 is still present for firms within the same sector. This exercise shows that the aggregate results are not driven uniquely by firms who are potentially direct competitors in the product market. Thus, a story about product market demand spillovers is less likely to be the culprit.

A second explanation concerns the coworkers who remain at the current employer. Suppose a high-premium firm expands by recruiting an experienced member of another firm's team. The departure may reduce the productivity of the remaining workers, either because they lose a source of knowledge or because the team needs time to reorganize. Their wages could then fall even as their measured employment alternatives improve, which would also imply that our elasticity would not be capturing the true effect of improving alternative opportunities.

I examine the measured composition of the workplace by adding the leave-one-out average of the estimated worker wage effects of the focal worker's team, defined as workers in the same establishment and occupation in a given year. The bias-corrected estimate with this control is -0.103. It remains negative and is numerically close to the main bias-corrected estimate of -0.100, weakening an explanation based on changes in average

TABLE 5. Elasticity of $\overline{\hat{\zeta}}$, by product market

	(1)
$\hat{\gamma}_S$: Same Sector	-0.0786***
	(0.0109)
$\hat{\gamma}_D$: Different Sector	-0.1037***
	(0.0061)
Worker FE	✓
Firm FE	✓
Market-Year FE	✓
Split-Sample?	✓
Weighted?	✓
Observations	16,064,983

Note: Non-linear least squares results from estimating (13). Same sector denotes the leave-out average of firms within the same market and 4-digit industry. Different sector denotes the leave-out average of firms within the same market but in different 4-digit industries. Weighted refers to current employment weights used in the computation of the leave-out average of firm effects. *** Denotes statistical significance at the 0.01 level.

measured coworker composition.

4.6.2. Current pay and future progression

I now examine whether this negative current-wage relationship is paired with faster future progression.

A job can be attractive because of what it pays today and because of what it may bring tomorrow. A worker may accept a lower starting wage when she expects faster progression, whether through later renegotiation, promotion, or a move to another employer. Models with on-the-job search and bargaining provide a natural mechanism for this pattern (Postel-Vinay and Robin 2002; Cahuc, Postel-Vinay, and Robin 2006). When a worker receives an outside offer, her current employer can respond with a raise; otherwise, she can move to the new firm. Anticipating this future competition, the current firm backloads compensation: it pays less at the beginning of the employment relationship and raises pay as outside offers arrive and the worker's bargaining position improves. The markdown is larger when attractive offers are more likely, which connects current pay directly to the quality of other firms in the market. Appendix F.1 develops this wage-backloading channel using the sequential-bargaining model of Berger et al. (2024). In the wage equation, it is clear that alternative employers may negatively influence wages through this compensating differential due to future job-ladder climbs.

TABLE 6. Elasticity of $\overline{\hat{\zeta}}$, controlling for team composition

	(1)	(2)
$\hat{\gamma}$	-0.0730***	-0.1029***
	(0.0045)	(0.0063)
Worker FE	✓	✓
Firm FE	✓	✓
Market-Year FE	✓	✓
Average worker FE	✓	✓
Split-Sample?	–	✓
Weighted?	✓	✓
Observations	16,064,983	16,064,983

Note: Non-linear least squares results from estimating (6) including a control for the leave-one-out average of the worker fixed effects of the team (measured as workers in the same establishment, occupation and year). Weighted refers to current employment weights used in the computation of the leave-out average of firm effects. *** Denotes statistical significance at the 0.01 level.

Skills and information provide a complementary route. Employment may develop portable skills, reveal ability, or establish a signal that other employers can evaluate. Appendix F.2, drawing on Gualdani et al. (2026), shows how the future value of such gains can enter current wages. Better-paying alternatives could increase the return to skills or information acquired at the present employer, allowing part of that return to be priced into current compensation as a potentially negative compensating differential.

Tenure I first ask whether the negative relationship is concentrated near the start of an employment relationship, when the scope for future progression is greatest. To explore this, I estimate the following model:

$$w_{ifmt} = \mathbf{x}'_{ifmt}\boldsymbol{\beta} + \alpha_i + \theta_{mt} + \zeta_f + \sum_{k=1}^{4} \delta_k D^{(k)}_{ifmt} + \overline{\zeta}_{-fmt}\left(\gamma_0 + \sum_{k=1}^{4} \gamma_k D^{(k)}_{ifmt}\right) + \varepsilon_{ifmt}, \tag{14}$$

where $D^{(k)}_{ifmt} = \mathbb{1}[T_{ifmt} \in B_k]$, T_{ifmt} denotes the worker's tenure with the firm in months, $B_0 = [0, 12)$, $B_1 = [12, 36)$, $B_2 = [36, 60)$, $B_3 = [60, 120)$, and $B_4 = [120, 600]$. Workers with tenure below 12 months constitute the omitted group. Hence, γ_0 is the elasticity for this baseline group, while $\gamma_0 + \gamma_k$ is the elasticity for tenure bracket B_k, for $k = 1, \ldots, 4$. This specification allows one to test whether $|\gamma_0| > |\gamma_0 + \gamma_k|$, i.e., whether the elasticity is larger for workers with less than 12 months of tenure than for workers in each subsequent tenure bracket, as predicted by theory.

Figure 2 shows a sharply front-loaded relationship. The coefficients can be found in Table 14. For exposition purposes, the figure plots $\gamma_0 + \gamma_k$ for each tenure bracket,

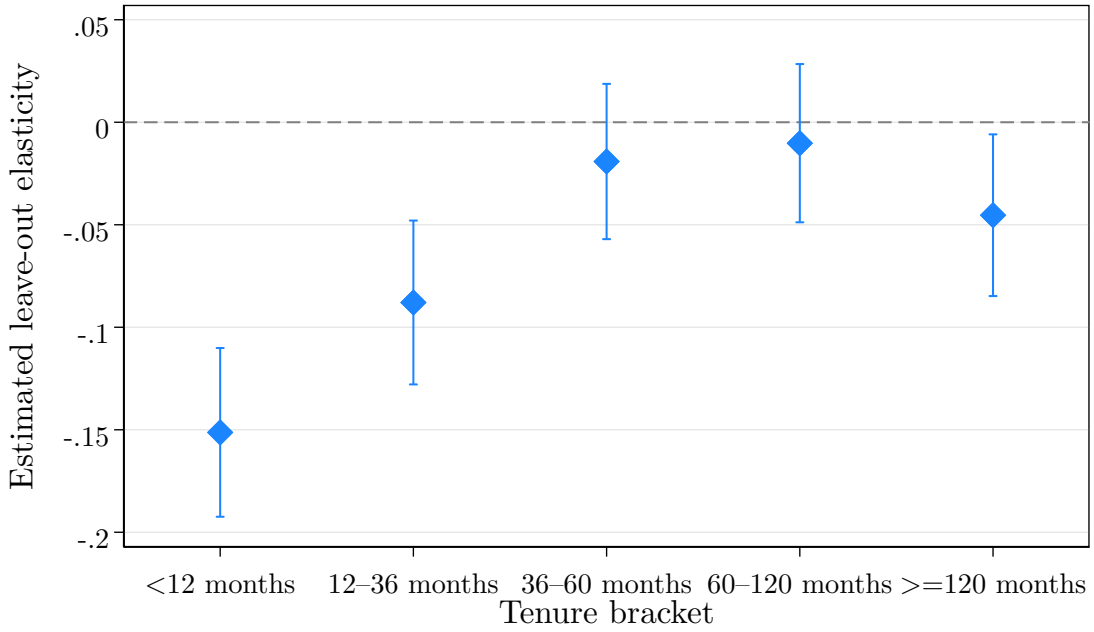


FIGURE 2. Elasticity of wages to other firms' pay premia, by tenure bracket

Notes: Coefficients from estimating equation (14). Each dot represents the term $\gamma_0 + \gamma_k, k = 1, \ldots, 4$. For workers with tenure < 12 months, it shows γ_0. 95% confidence intervals, clustered at the firm-year level.

alongside 95% confidence intervals. As anticipated, the effect of the leave-out average of firm effects is stronger for newly hired workers, and decreases with tenure. The elasticity is –0.151 during the first year and –0.088 between one and three years. Thus, a 10% increase in alternative-employer pay quality is associated with wages that are approximately 1.5% lower during the first year, compared with 0.9% lower between years one and three. The elasticity then falls to -0.019 and -0.01 between three and ten years of tenure, being statistically indistinguishable from zero. The central result is clear: the current-wage markdown is concentrated at the beginning of the employment relationship and attenuates strongly as tenure accumulates.

This profile is exactly what wage backloading predicts. Early in a match, future renegotiations and promotions still lie ahead, so the firm can tilt compensation toward the future. As tenure increases, workers have had time to obtain outside offers, renegotiate, and accumulate firm-specific knowledge. Their bargaining position strengthens, and the initial markdown largely unwinds.

Separations and promotions If better alternatives make future progression more valuable, they should predict both moves to other employers and advancement within the

TABLE 7. Elasticity of separations to outside options

	(1) LPM	(2) LPM	(3) LPM	(4) Cloglog	(5) Cloglog	(6) Cloglog
$\hat{\delta}_{\text{Separations}}$	0.0269***	0.0249***	0.0283***	0.0255***	0.0253***	0.0301***
	(0.0038)	(0.0034)	(0.0033)	(0.0036)	(0.0031)	(0.0031)
Elasticity	0.4337	0.4024	0.4566	0.4115	0.4077	0.4857
Centrality	✓	✓	✓	✓	✓	✓
Worker Wage FE	–	✓	✓	–	✓	✓
Firm Wage FE	–	✓	✓	–	✓	✓
Market-Year Wage FE	–	–	✓	–	–	✓
Split-Sample?	✓	✓	✓	✓	✓	✓
Weighted?	✓	✓	✓	✓	✓	✓
Observations	15,664,813	15,664,813	15,664,813	15,664,813	15,664,813	15,664,813

Note: The dependent variable is a binary variable which equals 1 if the worker is in another firm in the next year. Controls for age and its square, education, tenure and a male dummy. The wage fixed effects are obtained from the estimation of equation (6). The elasticities are obtained by dividing the coefficients on $\overline{\zeta}_{-fmt}$ by the average separation rate in the sample. Split-sample refers to generated regressor adjustment as in Dhaene and Jochmans (2015). See Section 2.3.3 for details. Weighted refers to current employment weights used in the computation of the leave-out average of firm effects. Standard errors clustered at firm × year level. *** Denotes statistical significance at the 0.01 level.

current firm. For each outcome $r \in \{\text{promotion}, \text{separation}\}$, I estimate

$$y^r_{ifm,t+1} = \mathbf{x}'_{ifmt}\boldsymbol{\beta} + \delta\overline{\hat{\zeta}}_{-fm,t} + \kappa \cdot \text{centrality}_i + \varepsilon_{ifm,t+1} \tag{15}$$

where $y^r_{ifm,t+1}$ equals one if the worker is promoted or joins another firm between t and $t+1$; $\mathbf{x}_{ifmt}$ includes age, age squared, a gender indicator, education and tenure; $\overline{\hat{\zeta}}_{-fm,t-1}$ is the leave-out weighted average of firm effects in worker i's market in $t-1$; centrality_i is an eigenvector centrality measure, aimed at capturing the fact that more central workers tend to move between jobs more often; and ε_{ifmt} captures the error term. Furthermore, I also include as controls the worker, firm and market-year wage fixed effects estimated from equation (6)[2]. The coefficient of interest is δ, which should capture the quasi-elasticity of the outcome variable to changes in the estimated leave-out average of firm effects. Given the relatively low prevalence of separations in the dataset, I estimate equation (15) as a linear probability model as well as a complementary log-log (cloglog) model, reporting average marginal effects for the latter. The cloglog formulation also has the added benefit of mapping to a discrete-time hazard model with proportional hazards.

Table 7 shows a positive response of employer transitions. When translated to an elasticity, the cloglog base model shows that a 10% increase in the average outside firm

[2]The inclusion of these fixed effects as controls is not without controversy, as it may be attenuating the composition bias while, at the same time, aggravating the generated regressor bias.

TABLE 8. Elasticity of promotions to outside options

	(1) LPM	(2) LPM	(3) LPM	(4) Cloglog	(5) Cloglog	(6) Cloglog
$\hat{\delta}_{\text{Promotions}}$	0.0367***	0.0181***	0.0258***	0.0378***	0.0188***	0.0238***
	(0.0057)	(0.0042)	(0.0042)	(0.0055)	(0.0042)	(0.0041)
Elasticity	1.4949	0.7368	1.0502	1.5416	0.7650	0.9681
Centrality	✓	✓	✓	✓	✓	✓
Worker Wage FE	–	✓	✓	–	✓	✓
Firm Wage FE	–	✓	✓	–	✓	✓
Market-Year Wage FE	–	–	✓	–	–	✓
Split-Sample?	✓	✓	✓	✓	✓	✓
Weighted?	✓	✓	✓	✓	✓	✓
Observations	15,664,813	15,664,813	15,664,813	15,664,813	15,664,813	15,664,813

Note: The dependent variable is a binary variable which equals 1 if the worker is promoted over the following year. Controls for age and its square, education, tenure and a male dummy. The wage fixed effects are obtained from the estimation of equation (6). The elasticities are obtained by dividing the coefficients on $\overline{\zeta}_{-fmt}$ by the average promotion share in the sample. Split-sample refers to generated regressor adjustment as in Dhaene and Jochmans (2015). See Section 2.3.3 for details. Weighted refers to current employment weights used in the computation of the leave-out average of firm effects. Standard errors are clustered at firm × year level. *** Denotes statistical significance at the 0.01 level.

quality in the market is associated with a 4% higher chance of moving to another job. The results are not very sensitive to the modeling choice, ranging from 0.4 to 0.5 in the cloglog model with all wage fixed effects included as controls.

This result corroborates similar findings in the literature. Caldwell and Harmon (2019) shows that a standard deviation increase in her outside options measure leads to a 15% higher chance of moving to another job. Jäger et al. (2024) show that, once workers are better informed about their possible outside options, they tend to correctly update their beliefs and adjust job search accordingly. While the authors do not have data on actual job moves, these two factors should contribute to increased mobility.

When it comes to promotions, the estimated elasticities are even higher. Table 8 shows that a 1% increase in the leave-out average of firm effects is associated with a 1.5% increase in the chance of being promoted next year, although the estimates vary much more across specifications. Including worker and firm wage fixed effects as controls attenuates the elasticity, dropping to 0.8. Regardless, the positive association is present for both outcomes. Together, the promotion and separation results show that some workers use improved alternatives to move, while others progress within their current firm.

Wage growth within firm Promotions measure one form of progression, but wage renegotiation need not involve a formal change in job title. I therefore follow workers hired at a baseline date, $t = 0$, who remain with the same employer, and relate alternative-

employer pay quality at hiring to cumulative log-wage growth over the following five years. I estimate

$$w_{i,t+h} - w_{it} = b_h \bar{\hat{\zeta}}_{it} + \mathbf{x}'_{it}\beta_h + \phi_{f,h} + \psi_{G_{it},h} + \lambda_{A_{it},h} + u_{i,t,h}, \qquad h = 1, \ldots, 5. \tag{16}$$

where ϕ_f denotes firm effects, $\psi_{G_{it}}$ market-year fixed effects and $\lambda_{A_{it}}$ age-year fixed effects. The latter controls for the fact wage growth may be different depending on the experience of the worker. Because the outcome is $\log(W_{i,t+h}/W_{it})$, b_h is the elasticity of the wage relative to its starting level with respect to the initial leave-out average of firm effects. It describes cumulative growth over h years. It is distinct from γ, which describes the relationship with current wage levels.

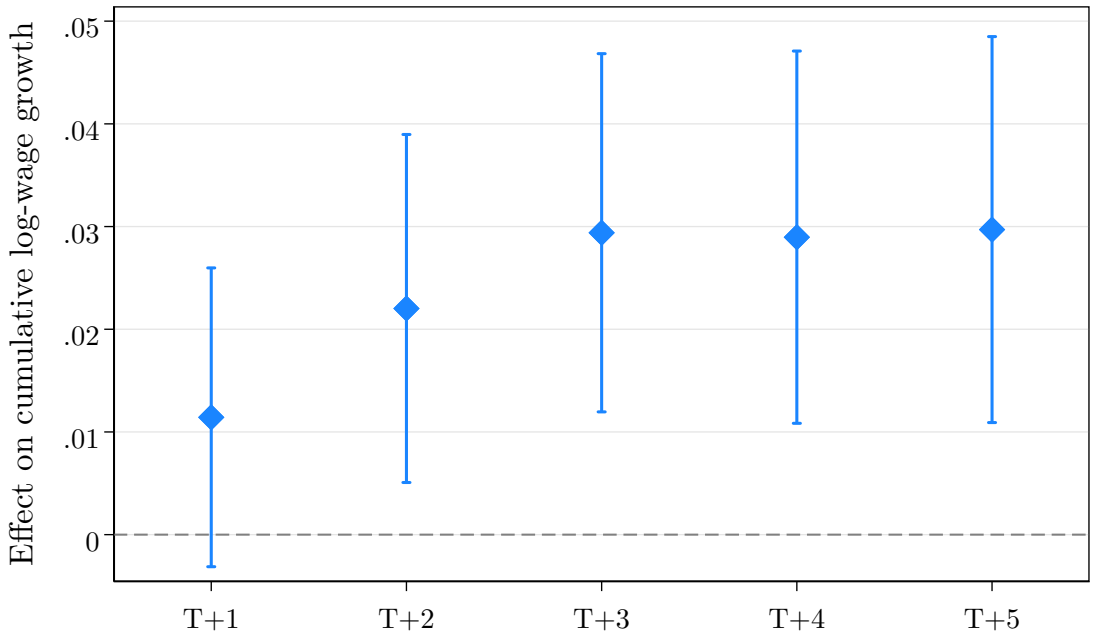


FIGURE 3. Alternative-employer pay quality and cumulative wage growth among newly hired stayers

Notes: Coefficients on the leave-out exposure at horizons one through five, with 95% confidence intervals, clustered at the firm-year level. The sample consists of newly hired workers who subsequently stay with the firm.

Figure 3 and Table 15 show that better alternatives at hiring predict faster wage growth within the firm. The elasticity rises from 0.011 after one year to 0.029 after three years and remains at approximately 0.030 through year five. In magnitude, a 10% increase in alternative-employer pay quality is associated with 0.11% greater cumulative wage growth after one year, 0.29% after three years, and 0.30% after five years.

The timing is informative. In Table 16 I provide the tests for the significance of pairwise differences between the coefficients. The three-year coefficient exceeds the one-year coefficient by 0.018 points ($p = 0.015$), and the five-year coefficient exceeds it by essentially the same amount ($p = 0.030$). By contrast, the three- and five-year coefficients are nearly identical ($p = 0.967$). Although the joint test does not reject equality across all five horizons ($p = 0.131$), the targeted contrasts show that the wage-growth advantage accumulates during the first three years and then stabilizes. This is the mirror image of the tenure result: the negative current-wage elasticity fades precisely when the positive wage-growth differential builds up.

Experience vs. tenure effects Can we distinguish between tenure and overall labor market experience effects? Figure 2 may conflate both channels, as long-tenured workers are also more experienced, by construction. Thus, these findings may simply suggest that those with more labor market experience have already reaped the benefits from on-the-job learning and human capital accumulation that younger workers still have not. However, Figure 3 provides evidence that, after controlling for age at hiring, higher outside options quality has a positive impact on wage growth, which is consistent with a story that goes beyond experience returns.

To investigate this, I estimate the main model separating experienced and early-career workers, as well as new hires and longer-tenured ones. Let D^1_{it} identify early-career recent hires, D^2_{it} early-career longer-tenured workers, and D^3_{it} experienced recent hires. Experienced longer-tenured workers are the reference group. The model is

$$w_{ifmt} = \mathbf{x}'_{it}\boldsymbol{\beta} + \alpha_i + \zeta_f + \theta_{mt} + \sum_{g=1}^{3} \delta_g D^g_{it} + \left(\gamma_0 + \sum_{g=1}^{3} \gamma_g D^g_{it}\right)\bar{\zeta}_{-fmt} + \varepsilon_{ifmt}. \tag{17}$$

The δ_g terms allow the groups to have different wage levels. The total elasticity to the leave-out firm effects average is γ_0 for experienced longer-tenured workers and $\gamma_0 + \gamma_g$ for each of the other groups. Thus, γ_g measures a difference from the reference group's elasticity, rather than a total elasticity on its own. $\gamma_1 - \gamma_2$ captures the effect of being a new hire for younger workers; $\gamma_1 - \gamma_3$ captures the experience effect; their difference, $(\gamma_1 - \gamma_2) - (\gamma_1 - \gamma_3) = \gamma_3 - \gamma_2$, indicates whether experience or tenure effects matter more for the negative elasticity.

The table shows that the negative elasticity is particularly strong among early-career new hires, exactly when future renegotiation, learning, and skill acquisition are most valuable. The effect of tenure is large, decreasing the elasticity by 0.14 points. The effect of experience is slightly smaller, but not negligible, at 0.12 points. This simple accounting exercise therefore shows that both dimensions matter for the negative elasticity.

Taken together, the results form a coherent dynamic-compensation pattern. Better alternative employers are associated with lower current pay primarily near hiring, higher

TABLE 9. Elasticity of $\bar{\hat{\zeta}}$ by tenure and experience status

	(1)
Long-tenure, experienced workers ($\hat{\gamma}_0$)	-0.0806***
	(0.0218)
Difference for:	
New hires, early-career workers ($\hat{\gamma}_1$)	-0.1996***
	(0.0109)
Long-tenure, early-career workers ($\hat{\gamma}_2$)	-0.0563***
	(0.0070)
New hires, experienced workers ($\hat{\gamma}_3$)	-0.0732***
	(0.0075)
Worker FE	✓
Firm FE	✓
Market-Year FE	✓
Split-Sample?	✓
Weighted?	✓
Observations	14,401,150

Note: Non-linear least squares results from estimating (17). New hires are workers with at most 24 months of tenure. Experienced workers are those with at least ten years of labor market experience. Weighted refers to current employment weights used in the computation of the leave-out average of firm effects. *** Denotes statistical significance at the 0.01 level.

probabilities of promotion and employer transition, and faster subsequent wage growth even among workers who remain with the same firm. The negative current-wage elasticity is therefore not evidence that better alternatives make workers worse off. It shows that compensation is tilted away from current pay and toward future progression. Search and renegotiation provide the central mechanism, while learning and skill acquisition reinforce it early in workers' careers.

These are non-exhaustive explanations. In particular, this paper is agnostic about the role of amenities and how these may impact our results. Recent work by Caldwell, Haegele, and Heining (2026) shows how amenities may relate to outside options and their importance in driving firm pay. Nonetheless, the evidence provided in this section is well aligned with those future-opportunity-based explanations.

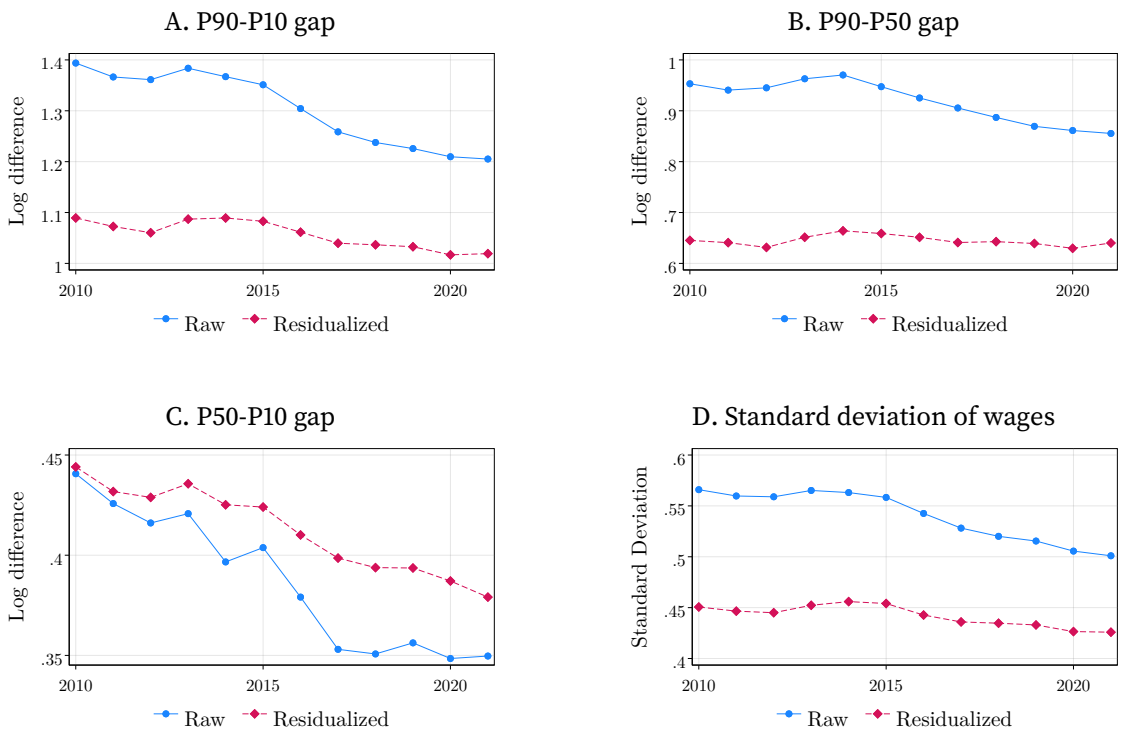


FIGURE 4. Measures of wage inequality, 2010–2021

Note: Log real hourly wages residualized according to a Mincer equation estimated for each year, controlling for quadratic terms in age and tenure and for education.

5. The impact of market pay structure on inequality

The study of the role of worker and firm unobserved heterogeneity in affecting wage inequality has developed substantially since the seminal work of Abowd, Kramarz, and Margolis (1999). By now, it is widely accepted that high-wage workers work for high-wage firms. Recent developments have tried to reconcile the statistical model of AKM with the wage-setting literature, bridging the gap between empirics and theory, trying to untangle why, in some cases, this worker-firm sorting is not as strong as a standard bargaining model could predict. Researchers have pointed to the role of occupation heterogeneity (Torres et al. 2018), the effect of previous firms' policies (Di Addario et al. 2023), learning about type (Gualdani et al. 2026), and even other forms of sorting (Lentz, Piyapromdee, and Robin 2023). The model in this paper adds another: what the labor market around that job looks like.

To perform this exercise, I do an AKM-style variance decomposition of log wages, where the variance of wages is decomposed into the contributions of worker, firm, market

structure and other time-varying controls:

(18)

$$
\begin{aligned}
\text{Var}\left[w_{ifmt}\right] &= \text{Var}\left[\mathbf{x}'_{ifmt}\boldsymbol{\beta} + \alpha_i + \theta_{mt} + \zeta_f + \gamma\overline{\zeta}_{-fmt} + \varepsilon_{ifmt}\right] \\
&= \text{Cov}\left[\mathbf{x}'_{ifmt}\boldsymbol{\beta}, w_{ifmt}\right] && \text{Contr. of covariates} \\
&+ \text{Cov}\left[\alpha_i, w_{ifmt}\right] && \text{Contr. of worker effects} \\
&+ \text{Cov}\left[\theta_{mt}, w_{ifmt}\right] && \text{Contr. of market-year effects} \\
&+ \text{Cov}\left[\zeta_f, w_{ifmt}\right] && \text{Contr. of firm effects} \\
&+ \text{Cov}\left[\gamma\overline{\zeta}_{-fmt}, w_{ifmt}\right] && \text{Contr. of leave-out firm effects} \\
&+ \text{Cov}\left[\varepsilon_{ifmt}, w_{ifmt}\right] && \text{Contr. of residual}
\end{aligned}
$$

where $\text{Cov}\left[\cdot, w_{ifmt}\right]$ represents the covariance between each element and the log of real hourly wages. It is important to see that, in this version of the decomposition, the covariance incorporates sorting. For example, the worker component is large not only when permanent worker wage differences are dispersed, but also when workers with high values of α_i are disproportionately employed in high-paying firms or markets. Moreover, a contribution can be negative when a component is systematically larger in observations whose other wage components are lower. Such a component compresses the wage distribution. I start with this version of the variance decomposition for exposition purposes and, in a second exercise, delve deeper into the covariance structure of the fixed effects.

For context, wage inequality in Portugal is pervasive but has been decreasing throughout the last decade. Figure 4 shows several measures of wage inequality, including percentile gaps and the standard deviation of wages. It can be seen that wage variance is still positive even after controlling for age, tenure, education and time, which suggests the need to account for unobserved factors such as ability, firm pay and market structure.

Figure 5 compares the standard two-way fixed effects (TFE, henceforth) model with the full model. For a more detailed version, see Table 17. In the TFE model, the contribution of worker characteristics to wage variance is close to 60%, whereas firms account for 26%. The remainder is split between covariates and unexplained factors. The contribution of worker heterogeneity to the variance is larger than that found in Torres et al. (2018) for Portugal, albeit using a different time period. After accounting for market structure effects, the component of the variance explained by workers is dampened to 52% and that of firms to 23%. These differences are offset by the contribution of the market-year fixed effects, which explain close to 15% of the wage variance. Market-year heterogeneity is therefore a first-order source of wage inequality: its contribution is almost two-thirds as large as the contribution of firms. Put differently, where a job sits in the labor-market network and when it is observed matter nearly as much for wage dispersion as the identity

of the employer itself.

The market-year component captures wage conditions shared by workers in the same market and year, including occupation-specific demand, local labor-market tightness, institutions, and shocks that affect the market as a whole. Its importance is not driven by a common national time trend. After residualizing the estimated market-year effects on year indicators, their contribution remains 11.8% of wage variance (Table 18). Market pay structure therefore explains persistent and economically meaningful wage differences across labor markets, beyond aggregate fluctuations. This result complements the evidence in Torres et al. (2018) that job-title heterogeneity is central to wage inequality in Portugal.

The leave-out firm component has a very different aggregate role. It contributes –1.1% of wage variance, which means that its observed covariance structure modestly compresses wages rather than widening their dispersion. This small contribution is fully consistent with the economically meaningful wage elasticity estimated earlier in the paper. The elasticity measures how an individual's wage responds when the pay quality of alternative employers changes; the variance decomposition also depends on how dispersed that exposure is and how it covaries with every other wage component. A meaningful marginal response can therefore coexist with a small aggregate contribution. The small effect in Figure 5 reveals that, while alternative employers matter for wage determination and worker mobility, persistent differences across workers, current firms, and market-year wage conditions generate almost all cross-sectional wage inequality. This pattern is consistent with the limited aggregate role of origin-firm pay policies in Di Addario et al. (2023) and of the outside-option channel in Berger et al. (2024).

Wage sorting In Appendix Section D.3, I show descriptive evidence that high-wage workers tend to work for high-wage firms and, to a lesser degree, in high-wage markets. It is also possible to dissect how much of this sorting contributes to the total variance of log wages. Table 10 provides the covariances between fixed effects and their respective contributions to the variance.

In the TFE model, the sorting of workers with firms explains 13% of the variance. This tells us that, in a counterfactual world where the sorting between workers and firms were random, inequality would decrease by 13%. This is a powerful, well-known result from the literature, which showcases that there is a strong preference for higher wages among high-ability individuals. The full model results show that this is not the only sorting mechanism that contributes to a larger wage dispersion. The fact that high-wage workers tend to sort into markets that pay more contributes to an increase in wage inequality by almost 9%. Hence, part of what appears to be sorting across employers in a standard AKM model is sorting across the markets in which those employers operate. Once markets are separated from firms, worker-firm sorting remains important, while worker-market

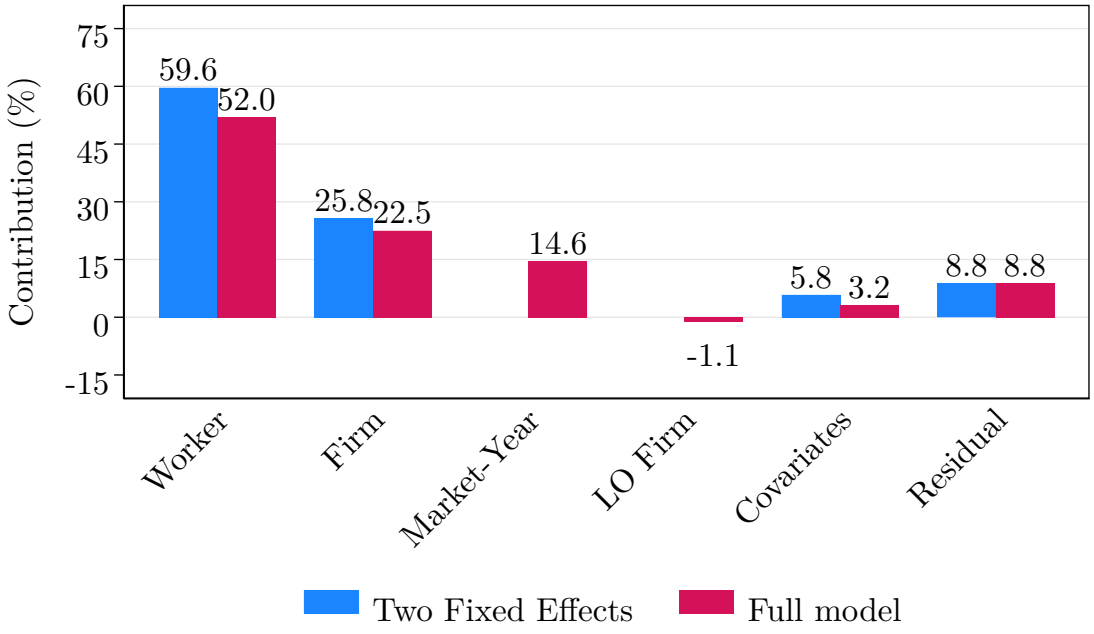


FIGURE 5. Share of wage variance explained by each component

Note: The contribution of each component is calculated according to (18). The worker fixed effects are adjusted for the linear effect of age and gender by residualizing $\hat{\alpha}_i$ on gender and year of birth dummies.

sorting is nearly as consequential for inequality. This can be explained by the fact that market-based amenities—such as place, industry or occupation—are also quite important for an individual's decision of where to work. Thus, when we observe a move to another firm, we need to account for the fact that the worker may, in fact, also be taking other market-specific variables into account in her decision. Lastly, as in the previous exercise, the share of variance explained by factors related to alternative firms is real, but quite modest.

Limited mobility bias Wage variance decompositions derived from estimated AKM-type models are subject to limited mobility bias, a form of incidental parameter bias that leads to underestimation of the contribution of the worker-firm sorting component (Bonhomme et al. 2023). Appendix G provides a more detailed analysis of the issue and shows that the aforementioned results are robust to the bias correction methods proposed by Andrews et al. (2008) and Kline, Saggio, and Sølvsten (2020).

6. Conclusion

In this paper, I investigate the role of other firms' pay policies and general market structure in determining wages. I propose a reduced-form equation where log wages depend not only

TABLE 10. Fixed effect covariances and contributions to Var $\left[w_{ifmt}\right]$

	Covariance	Share of total variance (%)
Two Fixed Effects		
Cov(Worker,Firm)	0.0384	13.07
Full Model		
Cov(Worker,Firm)	0.0310	10.50
Cov(Worker,Market-Year)	0.0256	8.71
Cov(Worker,LO Firm)	-0.0030	-1.04
Cov(Firm,Market-Year)	0.0068	2.33
Cov(Firm,LO Firm)	-0.0026	-0.91
Cov(Market-Year,LO Firm)	-0.0006	-0.22

Note: The full model shows the pairwise covariances among worker fixed effects, firm fixed effects, market-year fixed effects, and the leave-out average of firm fixed effects (LO Firm), with each covariance multiplied by 2. In both models, the worker fixed effects are adjusted for the linear effect of age and gender by residualizing $\hat{\alpha}_i$ on gender and year of birth dummies.

on worker, firm and market-year fixed effects, but also on the average of other firms' fixed effects. Drawing from the peer effects literature, I estimate a non-linear high-dimensional fixed effects model that jointly uncovers the distribution of fixed effects and the elasticity of wages to the leave-out average of other firms' fixed effects.

The results establish that other firms' pay policies matter for wage determination. The leave-out average of firm effects, $\overline{\zeta}_{-fmt}$, strongly predicts the premium of the firm to which a worker subsequently moves, confirming that it captures the quality of relevant employment alternatives. I find that the quality of outside options is positively associated with wages, showing that a 10% increase in the average quality of other firms in the same market leads to a 3% increase in wages. However, after controlling for common market shocks, idiosyncratic shocks to outside options have a general negative impact on wages. The sign reversal is robust to lagged and predicted employment weights and to broader definitions of the relevant labor market.

The negative conditional elasticity reflects the timing of compensation, as it is strongest at the beginning of the employment relationship and of one's career. I posit that the current-wage markdown and subsequent progression are two parts of the same wage-setting process: firms backload compensation when workers face valuable future opportunities, and workers realize that value through renegotiation, promotion, or movement to another employer.

Lastly, I investigate how market structure can impact wage inequality. An extension to the AKM variance decomposition framework shows that market-year effects explain close to 15% of wage variance in Portugal from 2010 to 2021. By contrast, the direct leave-out

firm component contributes –1.1% of wage variance and therefore modestly compresses the wage distribution. Thus, while alternative employers have a meaningful effect on individual wages and mobility, they are not important determinants of overall wage inequality.

These findings reinforce the fact that researchers should not neglect market structure when analyzing labor markets. Wages are determined not solely based on the policies of the current firm since other firms matter. Furthermore, they highlight how reduced-form models may cast aside relevant search-and-bargaining mechanisms if those mechanisms are not properly accounted fo

Further research could incorporate other types of data to more solidly understand empirically how outside options can impact wages through search and renegotiation. Recent advancements include the work by Jäger et al. (2024) on improving information about outside options and the use of data on accepted and rejected offers by Guo (2025a,b). Understanding how changes in the pay quality of employers are related to amenities (Caldwell, Haegele, and Heining 2026) is also a promising avenue for research. In general, further evidence on this matter could provide guidance on policies regarding pay transparency across firms, improving job-search behavior and reducing labor market power.

A. Auxiliary figures and tables

TABLE 11. Summary statistcs

	Mean	SD	P10	P50	P90
Wages					
Log real hourly wage	0.55	0.55	-0.01	0.39	1.32
Demographics					
Age	40.38	10.74	26.00	40.00	55.00
Share of Females	0.46	0.50	–	–	–
Education					
Less than High School	0.51	0.50	–	–	–
High School	0.31	0.46	–	–	–
College Degree	0.18	0.38	–	–	–
Employment					
Tenure (months)	103.39	107.09	5.00	64.00	263.00
Less than 10 Employees	0.25	0.43	–	–	–
10 to 50 Employees	0.26	0.44	–	–	–
50 to 250 Employees	0.22	0.41	–	–	–
More than 250 Employees	0.27	0.44	–	–	–

TABLE 12. Elasticity of wages to other firms' pay premia, all coefficients

	(1)	(2)	(3)	(4)
Age sq.	0.0002***	0.0002***	-0.0003***	-0.0003***
	(0.0000)	(0.0000)	(0.0000)	(0.0000)
Tenure	0.0014***	0.0016***	0.0007***	0.0008***
	(0.0000)	(0.0000)	(0.0000)	(0.0000)
Tenure sq.	-0.0000***	-0.0000***	-0.0000***	-0.0000***
	(0.0000)	(0.0000)	(0.0000)	(0.0000)
Education	0.0043***	0.0056***	0.0020***	0.0025***
	(0.0001)	(0.0002)	(0.0001)	(0.0002)
$\overline{\zeta}_{-fgt}$	0.2117***	0.2784***	-0.0731***	-0.1002***
	(0.0038)	(0.0073)	(0.0045)	(0.0063)
Worker FE	✓	✓	✓	✓
Firm FE	✓	✓	✓	✓
Market-Year FE	–	–	✓	✓
Split-Sample?	–	✓	–	✓
Weighted?	✓	✓	✓	✓
Observations	16,064,983	16,064,983	16,064,983	16,064,983

Note: Split-sample refers to incidental parameter bias adjusment as in Dhaene and Jochmans (2015). See Section 2.3.3 for details. Weighted refers to current employment weights used in the computation of the leave-out average of firm effects.
*** Denotes statistical significance at the 0.01 level.

TABLE 13. Elasticity of wages to other firms' pay premia, alternative peer groups

	Peer group definition	
	SBM only	Occ. × Location × Year
	(1)	(2)
$\hat{\gamma}$	-0.2692***	-0.1263***
	(0.0491)	(0.0169)
Worker FE	✓	✓
Firm FE	✓	✓
Market-Year FE	✓	✓
Split-Sample?	–	–
Weighted?	✓	✓
Observations	16,064,983	16,064,983

Note: Column 1 shows the result from estimating model (6) with peer group defined solely from the SBM, i.e., without restricting to occupation. In column 2, peer groups are defined as individuals in the same occupation, NUTS II and year. Split-sample refers to incidental parameter bias adjusment as in Dhaene and Jochmans (2015). See Section 2.3.3 for details. Weighted refers to current employment weights used in the computation of the leave-out average of firm effects. Standard errors are clustered at firm × year level.
*** Denotes statistical significance at the 0.01 level.

TABLE 14. Elasticity of wages to other firms' pay premia, by new hire status

	(1)
$\hat{\gamma}_0$: Tenure < 12 months	-0.1513***
	(0.0210)
$\hat{\gamma}_1$: Tenure 12–36 months	0.0634***
	(0.0084)
$\hat{\gamma}_2$: Tenure 36–60 months	0.1321***
	(0.0108)
$\hat{\gamma}_3$: Tenure 60–120 months	0.1411***
	(0.0139)
$\hat{\gamma}_4$: Tenure ≥ 120 months	0.1059***
	(0.0151)
Worker FE	✓
Firm FE	✓
Market-Year FE	✓
Split-Sample?	✓
Weighted?	✓
Observations	14,401,150

Note: Regression estimated with a tenure bracket indicators interacted with $\overline{\zeta}_{-fmt}$. Controls for age squared, tenure, tenure squared, education and a bracket indicator. Split-sample refers to incidental parameter bias adjusment as in Dhaene and Jochmans (2015). See Section 2.3.3 for details. Weighted refers to current employment weights used in the computation of the leave-out average of firm effects.
*** Denotes statistical significance at the 0.01 level.

TABLE 15. Cumulative wage growth among newly hired stayers

	(1)
$\hat{b}_1$: $t+1$	0.0114*
	(0.0069)
$\hat{b}_2$: $t+2$	0.0220***
	(0.0080)
$\hat{b}_3$: $t+3$	0.0294***
	(0.0083)
$\hat{b}_4$: $t+4$	0.0290***
	(0.0086)
$\hat{b}_5$: $t+5$	0.0297***
	(0.0089)
Firm FE	✓
Market-Year FE	✓
Age-Year FE	✓
Observations	1,017,420

Note: Results obtained from estimating equation (16). Includes controls for education and estimated wage worker fixed effects. Split-sample refers to incidental parameter bias adjusment as in Dhaene and Jochmans (2015). See Section 2.3.3 for details. Weighted indicates whether the average is weighted by market employment shares.
* Denotes statistical significance at the 0.1 level. *** Denotes statistical significance at the 0.01 level.

TABLE 16. Testing equality of coefficients from equation (16)

Null hypothesis	Difference	Std. err.	Statistic	p-value
$b_1 = b_2 = b_3 = b_4 = b_5$	—	—	$F = 1.78$	0.1305
$b_3 - b_1 = 0$	0.0180	0.0074	$t = 2.43$	0.015
$b_5 - b_1 = 0$	0.0183	0.0084	$t = 2.16$	0.030
$b_5 - b_3 = 0$	0.0003	0.0075	$t = 0.04$	0.967

Note: Standard errors clustered at the firm-year level. Joint test: $F(4, 74205)$; pairwise tests: $t(74205)$, two-sided.

TABLE 17. Variance decomposition, 2010-2021

	Variance Components	Contributions (%)
Var(Wages)	0.2942	
Worker		
Var(Worker)	0.1236	42.01
Cov(Worker,Firm)	0.0155	5.27
Cov(Worker,Market)	0.0128	4.35
Cov(Worker,OO)	-0.0015	-0.51
Cov(Worker,X)	0.0002	0.07
Firm		
Cov(Firm,Worker)	0.0155	5.27
Var(Firm)	0.0450	15.30
Cov(Firm,Market)	0.0034	1.16
Cov(Firm,OO)	-0.0013	-0.44
Cov(Firm,X)	-0.0007	-0.24
Market		
Cov(Market,Worker)	0.0128	4.35
Cov(Market,Firm)	0.0034	1.16
Var(Market)	0.0445	15.13
Cov(Market,OO)	-0.0003	-0.1
Cov(Market,X)	-0.0020	-0.68
Outside Options		
Cov(OO,Worker)	-0.0015	-0.51
Cov(OO,Firm)	-0.0013	-0.44
Cov(OO,Market)	-0.0003	-0.10
Var(OO)	0.0002	-0.07
Cov(OO,X)	-0.0001	-0.03
Covariates	0.0094	3.19
Residual	0.0260	8.85

Note: The worker fixed effects are adjusted for the linear effect of age and gender by residualizing $\hat{\alpha}_i$ on gender and year of birth dummies.

TABLE 18. Variance decomposition, adjusted to time trends

	Covariance	Share of total variance (%)
Var(y)	0.29	1.00
Cov(Adjusted Worker FE, y)	0.15	52.03
Cov(Firm FE, y)	0.07	22.46
Cov(Adjusted Market FE ,y)	0.03	11.81
Cov(LO Firm, y)	-0.00	-1.09
Cov(X,y)	0.02	5.94
Cov(Resid,y)	0.03	8.85

Note: The worker fixed effects are adjusted for the linear effect of age and gender by residualizing $\hat{\alpha}_i$ on gender and year of birth dummies. The market-year fixed effects are adjusted by residualizing the fixed effects on year dummies.

B. Microcanonical formulation of the stochastic block model

Section 2.1 presents the canonical degree-corrected stochastic block model because it offers an intuitive representation of the job mobility network. In that formulation, the number of transitions between two jobs is Poisson distributed, and the expected number of transitions depends on origin-specific outflows, destination-specific inflows, and the intensity of flows between their markets. However, in the estimation, I use the corresponding microcanonical degree-corrected stochastic block model (Peixoto 2017). The two formulations separate the same sources of variation, but differ in whether job degrees and block-to-block flows are matched in expectation or fixed at their observed values.

To describe the microcanonical formulation, let

$$k_p^- = \sum_q A_{pq} \qquad \text{and} \qquad k_p^+ = \sum_q A_{qp} \tag{19}$$

denote the observed out-degree and in-degree of job p, respectively. In the job mobility network, k_p^- is the total number of transitions originating in job p, while k_p^+ is the total number of transitions arriving at that job. Let

$$e_{uv} = \sum_{p:z_p=u} \sum_{q:z_q=v} A_{pq} \tag{20}$$

be the total number of observed transitions from market u to market v. The corresponding total outflows and inflows for market u are

$$e_u^- = \sum_v e_{uv} \qquad \text{and} \qquad e_u^+ = \sum_v e_{vu}. \tag{21}$$

Conditional on the market assignment $\mathbf{z}$, the microcanonical model treats the degree sequence $\mathbf{k} = \{k_p^-, k_p^+\}_{p=1}^J$ and the block-flow matrix $\mathbf{e} = \{e_{uv}\}_{u,v=1}^M$ as fixed. It then assigns probability to all directed multigraphs that reproduce these sufficient statistics. Under the multigraph convention, the probability of an adjacency matrix $\mathbf{A}$ is

$$P(\mathbf{A} \mid \mathbf{k}, \mathbf{e}, \mathbf{z}) = \frac{\prod_{u,v} e_{uv}! \prod_p k_p^-! k_p^+!}{\prod_u e_u^-! e_u^+! \prod_{p,q} A_{pq}!}. \tag{22}$$

In the observed network, $A_{pp} = 0$ because transitions that leave the worker in the same job are excluded. The likelihood nevertheless uses the conventional directed-multigraph

representation, under which repeated transitions between the same pair of jobs are represented as parallel edges.

Degree correction enters by conditioning on the observed in-degree and out-degree of each job. Consequently, a large establishment–occupation pair is allowed to generate many more transitions than a small one without being assigned to a different market for that reason alone. Conditional on these degree differences, the partition is informative about which origin and destination jobs are connected more frequently than would be expected from their total mobility alone.

The partition and the number of markets are selected using a minimum-description-length criterion. Let $P(\mathbf{z})$, $P(\mathbf{e} \mid \mathbf{z})$, and $P(\mathbf{k} \mid \mathbf{e}, \mathbf{z})$ denote the priors over the partition, the block-flow matrix, and the degree sequence. The description length can be written schematically as

$$\begin{aligned} \Sigma = \; & -\log P\left(\mathbf{A} \mid \mathbf{k}, \mathbf{e}, \mathbf{z}\right) - \log P\left(\mathbf{k} \mid \mathbf{e}, \mathbf{z}\right) \\ & -\log P\left(\mathbf{e} \mid \mathbf{z}\right) - \log P\left(\mathbf{z}\right). \end{aligned} \tag{23}$$

The first term rewards partitions that describe the observed worker flows well. The remaining terms penalize partitions and parameter configurations that require a more complex description. Thus, splitting the network into additional markets is valuable only when the improvement in fit is sufficient to compensate for the additional model complexity. This provides a data-driven choice of M and avoids selecting a separate block for every job.

I minimize equation (23) using the non-nested degree-corrected block model in *graph-tool*.

The connection with the canonical model in Section 2.1 is straightforward. In the canonical formulation, γ_p^- and γ_p^+ govern the expected out-degree and in-degree of each job, while B_{uv} governs expected flows between markets. In the microcanonical formulation, these expected quantities are replaced by their realized counterparts, k_p^-, k_p^+, and e_{uv}.

C. Matrix representation and estimation

This appendix presents the matrix representation of equation (6) and the estimation steps described in Section 2.3.2. The specification without market-year effects is obtained by omitting the corresponding design matrix and parameter vector.

C.1. Matrix Form

Let n denote the number of worker-year observations in the retained sample, I the number of workers, F the number of firms, G the number of observed market-year groups, and K the number of covariates.

The $n \times 1$ vector $\mathbf{W}$ contains log wages, and $\mathbf{X}$ is the $n \times K$ covariate matrix. The worker, firm and market-year indicator matrices are $\mathbf{D}_w$, $\mathbf{D}_f$ and $\mathbf{D}_{mt}$, with dimensions $n \times I$, $n \times F$ and $n \times G$, respectively. Their associated parameter vectors are $\alpha \in \mathbb{R}^I$, $\zeta \in \mathbb{R}^F$ and $\theta \in \mathbb{R}^G$.

For observation r, let $f(r)$ denote its employer and $(m(r), t(r))$ its market-year group. Let $\mathcal{F}_{mt}$ contain the firms in that group, $F_{mt} = |\mathcal{F}_{mt}|$, and ω_{jmt} denote firm j's weight, with $\sum_{j \in \mathcal{F}_{mt}} \omega_{jmt} = 1$. In the main specification, these are employment shares.

Define the $n \times F$ matrix of leave-out firm weights, $\mathbf{P}$, by

$$P_{rj} = \begin{cases} \dfrac{\omega_{j,m(r),t(r)}}{1 - \omega_{f(r),m(r),t(r)}}, & j \in \mathcal{F}_{m(r),t(r)}, \quad j \neq f(r), \\ 0, & \text{otherwise.} \end{cases} \tag{24}$$

Each row sums to one. The vector $\mathbf{P}\zeta$ therefore contains the leave-out averages of firm premia. Exclusion is at the firm level, so other establishments belonging to the current employer do not enter its workers' alternative-employer index. Equal weighting across firms is obtained by setting $\omega_{jmt} = 1/F_{mt}$, which gives $P_{rj} = 1/(F_{mt} - 1)$ for each alternative firm.

The observation-level averaging matrix $\mathbf{V}$ provides an equivalent representation. Let c_{jmt} count the observations belonging to firm j in market-year mt, and assume that each positive-weight firm in the index is represented in that group. Define the $n \times n$ matrix by

$$V_{rs} = \begin{cases} \dfrac{\omega_{f(s),m(r),t(r)}}{c_{f(s),m(r),t(r)} \left(1 - \omega_{f(r),m(r),t(r)}\right)}, & \begin{array}{l} m(s) = m(r), \quad t(s) = t(r), \\ f(s) \neq f(r), \end{array} \\ 0, & \text{otherwise.} \end{cases} \tag{25}$$

Each firm's weight is distributed equally across its observations. Thus, $\mathbf{V}\mathbf{D}_f = \mathbf{P}$ and $\mathbf{V}\mathbf{D}_f\zeta$ produces exactly the same leave-out index.

After ordering observations by market-year, $\mathbf{V}$ is block diagonal. Every entry connecting workers at the same employer is zero, including workers at different establishments of that firm. The sets of firms can overlap across markets even though each observation belongs to one market-year block. The matrix is generally asymmetric, so its transpose must be retained in the estimation equations.

The wage equation can now be written as

$$\mathbf{W} = \mathbf{X}\boldsymbol{\beta} + \mathbf{D}_w\boldsymbol{\alpha} + \mathbf{D}_{mt}\theta + (\mathbf{I}_n + \gamma\mathbf{V})\,\mathbf{D}_f\zeta + \varepsilon, \tag{26}$$

$$= \mathbf{X}\boldsymbol{\beta} + \mathbf{D}_w\boldsymbol{\alpha} + \mathbf{D}_{mt}\theta + \left(\mathbf{D}_f + \gamma\mathbf{P}\right)\zeta + \varepsilon. \tag{27}$$

C.1.1. Toy example

To better visualize (26), consider the following example with 6 workers, 3 firms, 2 markets and only one time-period. For simplicity, let us ignore the covariates and market matrices. In this example consider that

a. Workers 1 and 2 work in Firm 1 and belong to the same market (M_1);
b. Worker 3 is in Firm 2 and belongs to market M_1;
c. Workers 4 and 6 are in Firm 3 and in market M_2;
d. Worker 5 is in Firm 2 but in market M_2;

Thus, in this example, market M_1 includes firms 1 and 2, and market M_2 includes firms 2 and 3. Hence, Firm 2 spans two different markets, so peer groups are not disjoint. In this case, the matrices would be

$$\begin{bmatrix} w_1 \\ w_2 \\ w_3 \\ w_4 \\ w_5 \\ w_6 \end{bmatrix} = \underbrace{\begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \end{bmatrix}}_{\mathbf{D_w}} \begin{bmatrix} \alpha_1 \\ \alpha_2 \\ \alpha_3 \\ \alpha_4 \\ \alpha_5 \\ \alpha_6 \end{bmatrix} + \left(\gamma \underbrace{\begin{bmatrix} 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 \\ 1/2 & 1/2 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1/2 & 0 & 1/2 \\ 0 & 0 & 0 & 0 & 1 & 0 \end{bmatrix}}_{\mathbf{V}} + \begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \end{bmatrix} \right) \underbrace{\begin{bmatrix} 1 & 0 & 0 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}}_{\mathbf{D_f}} \begin{bmatrix} \zeta_1 \\ \zeta_2 \\ \zeta_3 \end{bmatrix} + \varepsilon$$

C.2. Estimation

Write $\mathbf{A}_\gamma = \mathbf{D}_f + \gamma\mathbf{P} = (\mathbf{I}_n + \gamma\mathbf{V})\mathbf{D}_f$. The residual vector is

$$\mathbf{e} = \mathbf{W} - \mathbf{X}\boldsymbol{\beta} - \mathbf{D}_w\boldsymbol{\alpha} - \mathbf{D}_{mt}\theta - \mathbf{A}_\gamma\zeta. \tag{28}$$

The nonlinear least-squares estimates solve

$$\min_{\beta,\alpha,\theta,\gamma,\zeta} \mathbf{e}'\mathbf{e}, \tag{29}$$

with the fixed-effect normalizations imposed. Let $\hat{\mathbf{e}}$ denote the residual at the estimated parameters. The first-order conditions are

$$\mathbf{X}'\hat{\mathbf{e}} = \mathbf{0}, \tag{30}$$

$$\mathbf{D}_w'\hat{\mathbf{e}} = \mathbf{0}, \tag{31}$$

$$\mathbf{D}_{mt}'\hat{\mathbf{e}} = \mathbf{0}, \tag{32}$$

$$(\mathbf{P}\hat{\zeta})'\hat{\mathbf{e}} = \hat{\zeta}'\mathbf{D}_f'\mathbf{V}'\hat{\mathbf{e}} = 0, \tag{33}$$

$$\mathbf{A}_{\hat{\gamma}}'\hat{\mathbf{e}} = \mathbf{D}_f'(\mathbf{I}_n + \hat{\gamma}\mathbf{V}')\hat{\mathbf{e}} = \mathbf{0}. \tag{34}$$

Following Arcidiacono et al. (2012) and Portugal et al. (2025), these conditions can be implemented through alternating linear estimation steps. Conditional on $\hat{\gamma}$ and the other wage components, the firm-effect condition becomes

$$\mathbf{A}_{\hat{\gamma}}'\mathbf{A}_{\hat{\gamma}}\hat{\zeta} = \mathbf{A}_{\hat{\gamma}}'\left(\mathbf{W} - \mathbf{X}\hat{\beta} - \mathbf{D}_w\hat{\alpha} - \mathbf{D}_{mt}\hat{\theta}\right). \tag{35}$$

In the original notation, the matrix on the left is $\mathbf{D}_f'(\mathbf{I}_n + \hat{\gamma}\mathbf{V}')(\mathbf{I}_n + \hat{\gamma}\mathbf{V})\mathbf{D}_f$. This is a linear system in the firm premia when $\hat{\gamma}$ is held fixed.

An initial vector $\hat{\zeta}_0$ can be obtained from the additive wage model with $\gamma = 0$. At iteration h, the updates are:

a. Given $\hat{\zeta}_{h-1}$, estimate $\hat{\beta}_h$, $\hat{\alpha}_h$, $\hat{\theta}_h$ and $\hat{\gamma}_h$ by OLS in

$$\mathbf{W} - \mathbf{D}_f\hat{\zeta}_{h-1} = \mathbf{X}\beta + \mathbf{D}_w\alpha + \mathbf{D}_{mt}\theta + \gamma\mathbf{P}\hat{\zeta}_{h-1} + \mathbf{u}. \tag{36}$$

b. Given these estimates, define $\mathbf{A}_h = \mathbf{D}_f + \hat{\gamma}_h\mathbf{P}$ and update the firm premia by solving

$$\mathbf{A}_h'\mathbf{A}_h\hat{\zeta}_h = \mathbf{A}_h'\left(\mathbf{W} - \mathbf{X}\hat{\beta}_h - \mathbf{D}_w\hat{\alpha}_h - \mathbf{D}_{mt}\hat{\theta}_h\right). \tag{37}$$

   The chosen fixed-effect normalizations are maintained when expressing the updated wage components.

c. Repeat the two updates until the stopping criterion is met.

The calculations do not require constructing the $n \times n$ matrix $\mathbf{V}$. Multiplication by $\mathbf{P}$ and $\mathbf{P}'$ can be implemented using firm–market-year totals and observation-level aggregation.

Together with sparse indicator matrices, these operations permit iterative linear solvers for equation (37) and absorption of the worker and market-year effects in equation (36).

C.3. Inference

The linearization of the fitted wage with respect to all parameters has Jacobian

$$\hat{\mathbf{J}} = \left[\begin{array}{cccc} \mathbf{X} & \mathbf{D}_w & \mathbf{D}_{mt} & \mathbf{D}_f + \hat{\gamma}\mathbf{P} \quad \mathbf{P}\hat{\zeta} \end{array} \right]. \tag{38}$$

The firm-premium derivative contains both the own-firm indicator and the firm's contribution to alternative-employer exposure. The derivative with respect to γ is the estimated leave-out index. Thus, both channels enter the joint nonlinear least-squares calculation.

Following the Gauss–Newton approach described in Appendix B of Portugal et al. (2025), inference can be obtained from an auxiliary regression of the fitted residuals on the columns of $\hat{\mathbf{J}}$. For inference on the common slopes, the high-dimensional columns can be partialled out. Standard errors are clustered at the firm-year level.

D. Firm and market effects: additional descriptive results

This appendix presents the distributions, characteristics and dynamics of the firm and market-year effects discussed briefly in Section 4.2. It also describes worker sorting across firms and markets. These results provide context for the wage estimates and the inequality decomposition, without assigning a particular economic mechanism to either fixed effect.

D.1. Distributions and market characteristics

Figure 6 shows the distributions of $\hat{\zeta}$ and $\hat{\theta}$, i.e., the firm and market-year effects, respectively. Given that these effects are identified up to a constant, I normalize the firm effects such that the average of the Accommodation and Food Services industry is zero, while the market-year effects are normalized such that the average of the 2010 effects is zero. The figure shows a rather symmetric firm-effects distribution that closely mimics a Gaussian with the same mean and standard deviation. There is a higher concentration of firms around the mean relative to the Gaussian, but the distribution is otherwise similar. The market-year effects have a longer right tail and are mostly positive relative to the 2010 average. Their pooled distribution combines differences between markets with changes over time.

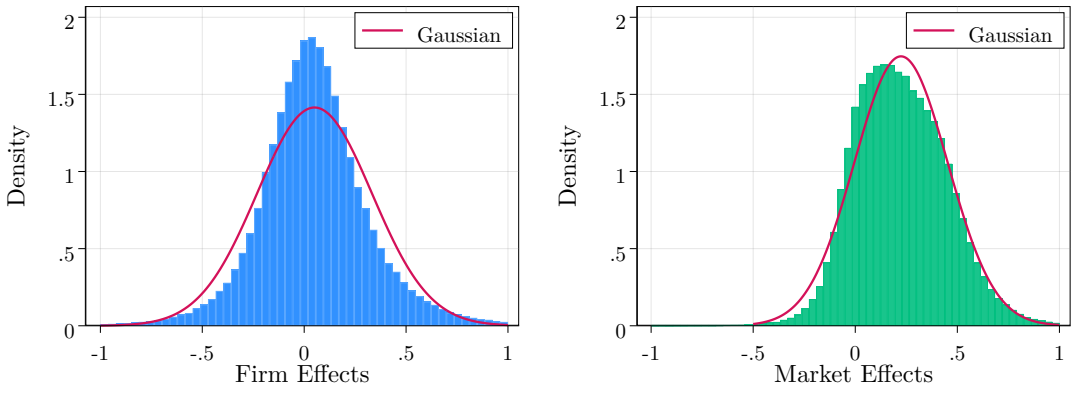


FIGURE 6. Fixed effects distributions

Notes: Firm effects are normalized such that their average in Accommodation and Food Services is zero. Market-year effects are normalized such that their average in 2010 is zero.

Figure 7 presents average market-year effects by occupation, industry and location. Managers, skilled professionals, and technicians and associate professionals have larger market premia than the remaining occupation groups. When looking at industries, information and communication, finance, real estate, and professional, scientific and technical activities present the largest premia, while Accommodation and Food Services shows the lowest. As for location, proxied by NUTS II, market premia are higher in the Lisbon area, followed by the North and Center, with Alentejo and Algarve at the bottom.

The differences across occupations are particularly pronounced. This is consistent with the importance of occupational pay differences documented by Torres et al. (2018), who show that one fifth of wage variation in Portugal is due to variation in job-title premia.

D.2. Do market effects drift?

Given the time dimension captured in θ, a natural question arises: how do market effects evolve over time? Previous work has shown that firm effects have a relatively large autocorrelation, implying that firms' pay premia are rather sticky (Lachowska et al. 2023). Figure 8 describes the evolution of market effects for a balanced panel of markets observed in all twelve years of the sample. Panel A compares market effects in 2010 and 2021. In general, those in 2021 are larger than in 2010, with stronger deviations at the bottom and top of the 2010 distribution. Figure 9 presents the comparison by industry.

Panel B shows the trend for the average market, with 2010 equal to zero. Market effects show little improvement during the recession years at the beginning of the period. The positive trend picks up after 2013, alongside the recovery of the Portuguese economy. By 2021, the average market effect is about 0.5 log points higher than in 2010. This is an increase in the estimated common market-year wage component; it does not, by itself, distinguish the role of outside opportunities from other changes in market conditions.

D.3. Worker sorting across firms and markets

Another relevant question is how workers sort across firms and markets. From the inequality literature, much is known about worker-firm sorting. In general, high-wage workers tend to work for high-premium firms, although sorting can change as workers age or accumulate tenure (Lentz, Piyapromdee, and Robin 2023; Lamadon et al. 2024).

The left panel of Figure 10 shows a bivariate density plot, with worker fixed-effect deciles on the vertical axis and firm fixed-effect deciles on the horizontal axis. Each square's color represents the number of workers in that combination. A darker color indicates a larger number of workers. In line with previous studies, the figure shows positive sorting between high-wage workers and high-premium firms, particularly in the top two deciles of each distribution.

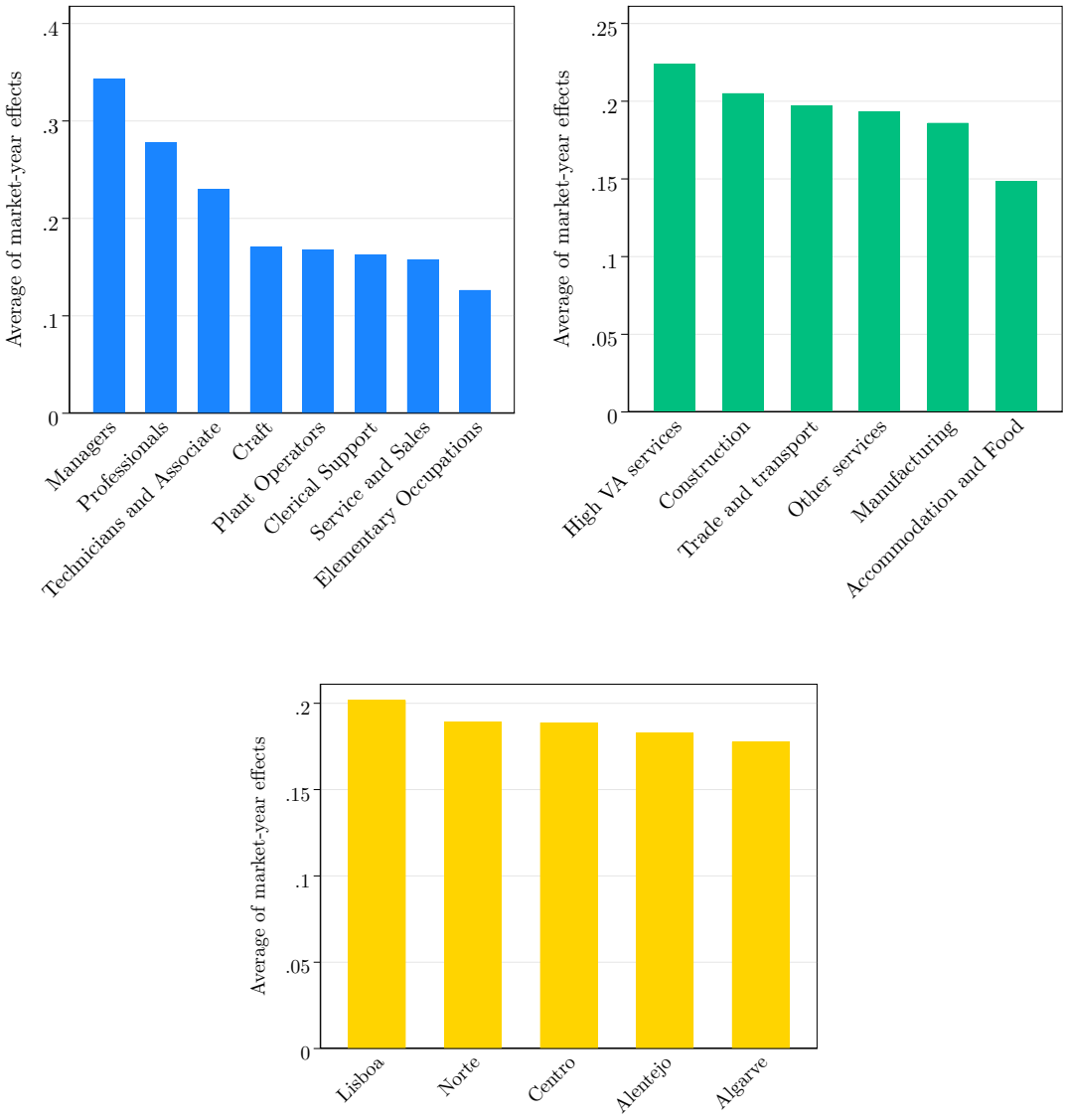
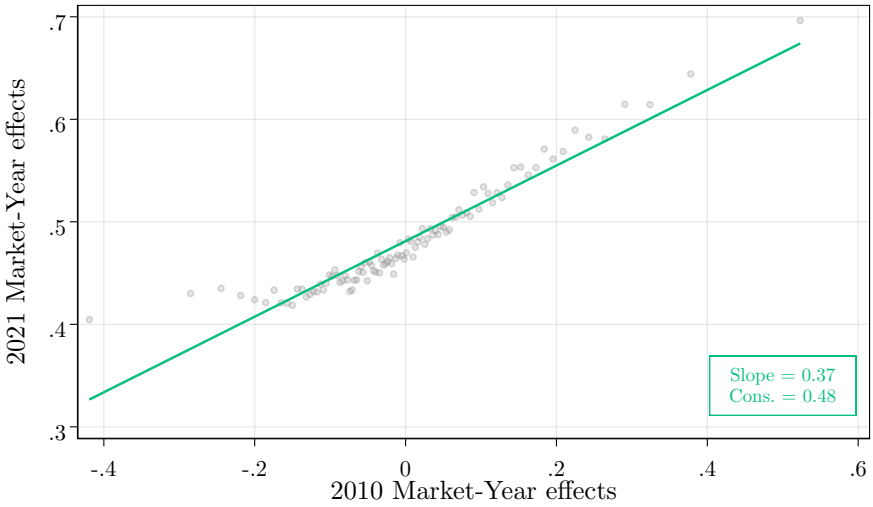
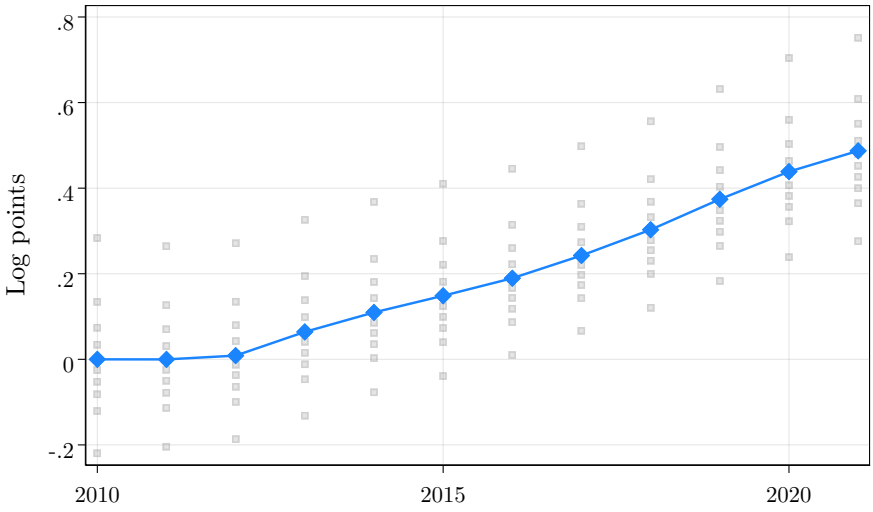


FIGURE 7. Average market-year fixed effects

Notes: The upper panels show average market-year effects by 1-digit ISCO occupation and industry or group of industries. The lower panel shows averages by NUTS II region. High VA services = Information and Communications; Financial Services; Real Estate Activities; Professional, Scientific and Technical Activities. Market-year effects are normalized such that their average in 2010 is zero.



A. 2010 vs. 2021



B. Average market effects over time

FIGURE 8. Dynamics of market effects

Notes: Balanced panel of markets observed in all twelve years of the sample. Market-year effects are normalized such that their average in 2010 is zero. Panel B shows average market effects weighted by market employment.

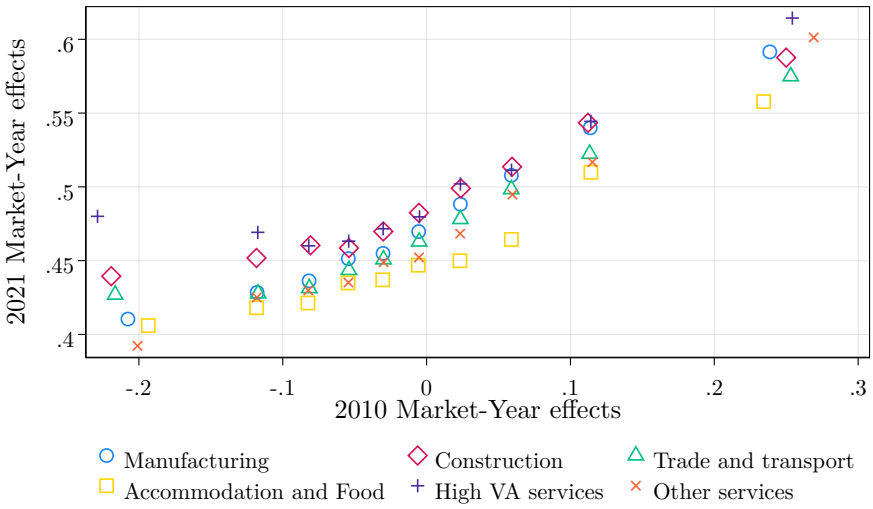


FIGURE 9. Market effects in 2010 and 2021, by industry

Notes: Market-year effects are normalized such that their average in 2010 is zero.

The right panel uses deciles of market-year effects averaged across years, with one average effect per market. The pattern is weaker than for worker-firm sorting, with much of the density around the median and some concentration in the upper-right corner. This comparison suggests that sorting across markets is less pronounced than sorting across firms when market effects are averaged over time.

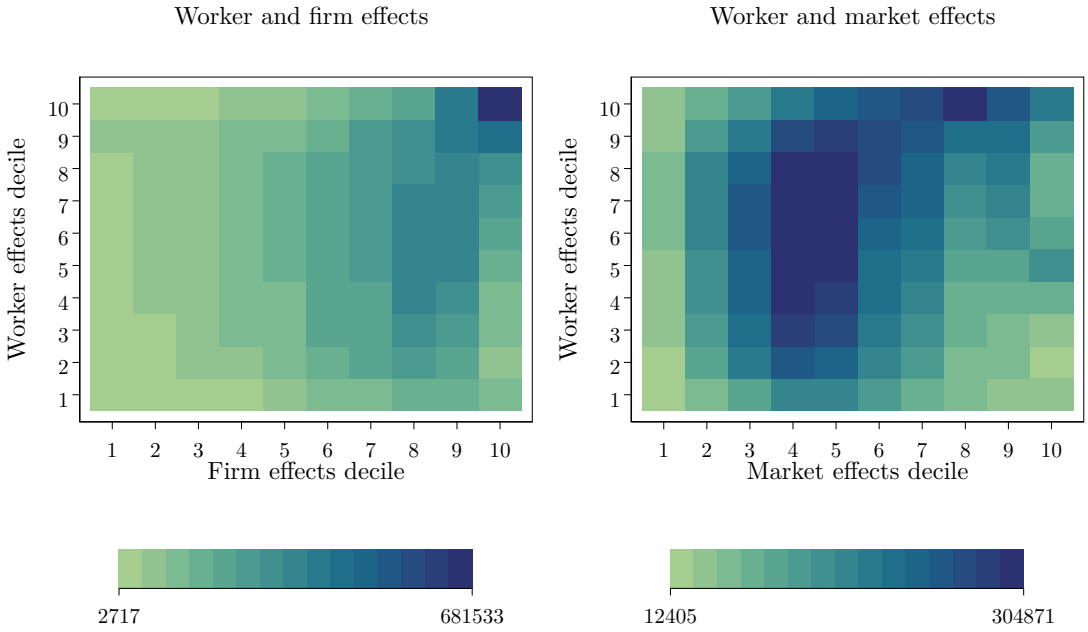


FIGURE 10. Correlation between fixed effects

Notes: The left panel compares worker and firm fixed-effect deciles. The right panel compares worker fixed-effect deciles and deciles of market-year effects averaged across years. Darker cells indicate more workers.

To provide further illustration, Figure 11 shows market-year effects over the twelve-year period by worker fixed-effect decile. Low-wage workers initially work in markets with higher estimated effects than their high-wage counterparts. This ranking reverses later in the sample. Higher-wage workers may be moving to markets with higher premia or working in markets whose premia grow faster. The figure does not distinguish these possibilities, and calendar time in the sample should not be interpreted as workers' accumulated experience. Section 5 examines the contribution of sorting to wage inequality.

D.4. Destination premia for all firm movers

Figure 12 extends the destination-premium comparison in Section 4.2 to workers who switch firms across markets as well as within them. The relationship remains positive, although it is weaker than for within-market movers. This is consistent with the leave-out

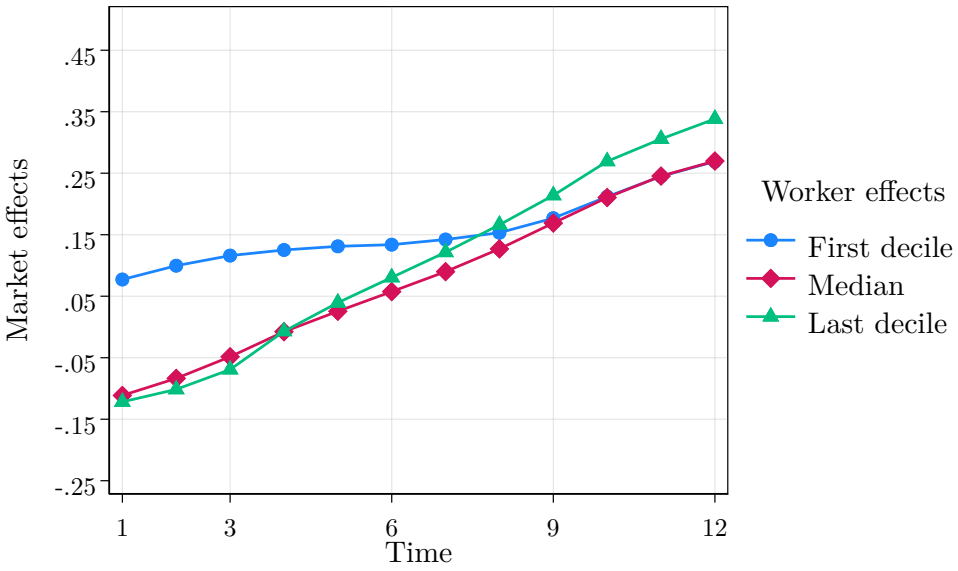


FIGURE 11. Market-year fixed effects over time, by worker fixed-effect decile

index summarizing alternative employers in the worker's origin market. The comparison concerns observed movers and does not recover offers that workers rejected or never received.

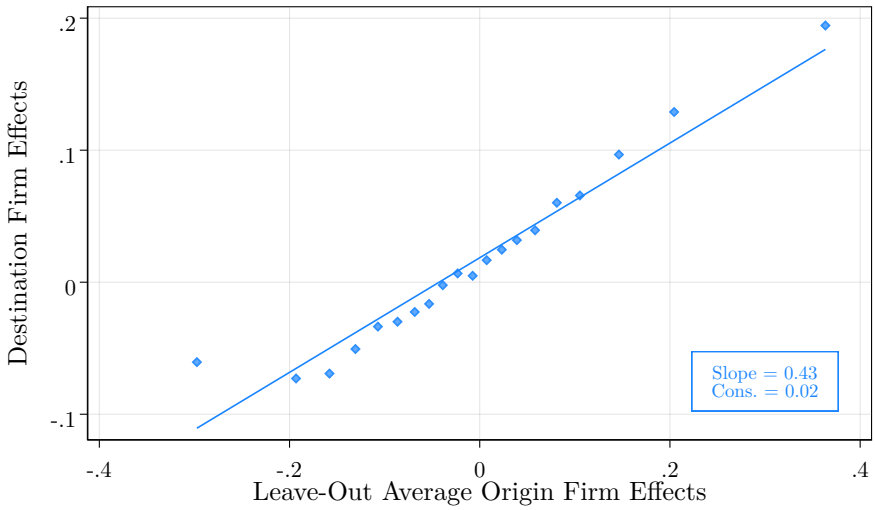


FIGURE 12. Destination firm effects and the leave-out average: all firm movers

Notes: Includes workers who change firms within and across markets. The leave-out average refers to the worker's origin firm and market in the previous year.

D.5. Detailed summary statistics

Table 19 extends Table 2 with the education, gender and collective bargaining splits, as well as skewness. Workers with a college degree face higher average alternative-firm premia than those without one, as do men relative to women and workers uncovered by a collective bargaining agreement relative to covered workers. As in the main text, these statistics describe the level of alternative employers' premia, rather than their difference from the worker's current firm. They do not establish why groups have different employment alternatives or how those differences affect their wages.

TABLE 19. Detailed summary statistics of leave-out average firm effects, by group

	Mean	Median	Skewness
Full sample	0.1771	0.1642	0.5376
Education			
No college	0.1596	0.1480	0.5389
College	0.2420	0.2244	0.3307
Gender			
Female	0.1598	0.1459	0.6990
Male	0.1922	0.1811	0.3953
CBA coverage			
No CBA	0.2114	0.2175	0.0613
CBA	0.1722	0.1572	0.6110
Wage quantiles			
Below P25	0.1088	0.1029	0.4710
P25 to P50	0.1482	0.1443	0.1796
P50 to P90	0.2072	0.2004	0.3589
Above P90	0.2997	0.2798	0.0539
Labor market concentration			
Low concentration	0.1405	0.1328	0.4428
High concentration	0.2138	0.2014	0.4654

Notes: Employment-weighted leave-out average of estimated firm effects in the same SBM market and occupation, excluding the worker's current firm and all its establishments. Means and medians are in log points; skewness is unitless. Firm effects are normalized such that their average in Accommodation and Food Services is zero. Low concentration = market-year HHI below the yearly median. CBA = collective bargaining agreement.

E. Heterogeneity

This section dissects the heterogeneity of the elasticity of wages to the leave-out average quality of firm pay across several dimensions: education, occupation groups, industries and level of labor market concentration. The first two can be seen as an exploration of the skill dimension. More educated workers and those in more cognitive-intensive occupations are, in the economics literature, usually referred to as high-skilled workers. While the inherent ability of the worker is averaged out in equation (6) due to the worker fixed effect, there is still value in assessing the differences across groups, as there may be elements of the market pay structure that interact with skill in a non-additive, time-varying manner (Bagger et al. 2014; Gualdani et al. 2026). Moreover, there is also recent evidence showing that information about wage offers may be quite heterogeneous across occupations (Batra, Michaud, and Mongey 2023), which evidently affects the channels investigated in this paper. The industry dimension is also interesting to investigate, as industries may vary in terms of pay transparency and mobility patterns through collectively bargaining agreements (de Almeida Vilares and Reis 2022) and non-compete agreements (Krueger and Ashenfelter 2022; Johnson, Lavetti, and Lipsitz 2025), both of which may also interact with job search. Lastly, performing a heterogeneity analysis by the level of labor market concentration speaks to recent advances in the monopsony literature, which show a direct link between higher concentration and lower wages through the outside options channel (Berger, Herkenhoff, and Mongey 2022; Jarosch, Nimczik, and Sorkin 2024; Berger et al. 2024).

Education Table 20 shows the coefficients for separate regressions by education level, where low-educated workers are those with at most twelve years of education, corresponding to high school in the Portuguese system. Whereas in the main model, one worker could have in their peer group workers from all education levels (provided that they worked in the same market and occupation), now peer groups are constrained to have workers in the same education group (within-group they may still have distinct years of schooling). The results show that only low-educated workers are affected by changes in the average quality of competitor firms in the market, after accounting for market-year fixed effects. For the less educated, a 10% increase in the average quality of competitors is associated with 1.3% lower wages, whereas for higher education workers, the results are indistinguishible from zero. This can be driven by several factors. First, more educated workers may have a higher bargaining power, therefore receiving a larger share of the match surplus. If this is true, then these workers are paid a wage that is closer to their marginal product, and firms are not able to mark down wages, rendering the wage backloading channel irrelevant for these workers. Another possible explanation hinges on the possibility that the wage ladder is steeper for low educated workers, thus leading to a higher wage markdown that

TABLE 20. Elasticity of wages to other firms' pay premia, by education group

	Low education		High education	
$\hat{\gamma}$	-0.1026***	-0.1338***	-0.0023	0.0002
	(0.0048)	(0.0069)	(0.0077)	(0.0122)
Worker FE	✓	✓	✓	✓
Firm FE	✓	✓	✓	✓
Market-Year FE	✓	✓	✓	✓
Split-Sample?	–	✓	–	✓
Weighted?	✓	✓	✓	✓
Observations	12,327,715	12,327,715	2,949,699	2,949,699

Note: Separate regressions for low and high education workers. Low education refers to workers that have a level lower than or equal to high school. Split-sample refers to incidental parameter bias adjusment as in Dhaene and Jochmans (2015). See Section 2.3.3 for details. Weighted refers to current employment weights used in the computation of the leave-out average of firm effects. Standard errors are clustered at firm × year level.
*** Denotes statistical significance at the 0.01 level.

insures against future raises.

Occupation Table 21 shows the results from estimating the model separately by occupation group. It shows that the elasticity of wages to the average quality of alternative firms is larger (more negative) for occupation groups that usually require less education and in which the type of work is more commonly routine or manual-intensive, such as sales, craft trades and elementary occupations. The elasticities reach up to -0.4 in craft and elementary occupations. Conversely, skilled and technical professionals have lower elasticities, ranging from -0.04 to -0.08. The estimate for managerial professionals is large but imprecise, being statistically indistinguishable from zero.

These results are in line with some of the literature on bargaining power estimation. In their seminal work, Cahuc, Postel-Vinay, and Robin (2006) estimate a larger bargaining power for more skilled occupations, even reaching the conclusion that, for the less skilled, the workers' share of the match surplus can be zero. From their model, such a result implies a strong effect from the quit/renegotiation channel. de Almeida Vilares and Reis (2022) estimate trends in bargaining power for different groups of workers based on their job title within collectively bargained agreements, in Portugal. The authors show that managers have a large degree of bargaining power, whereas skilled and unskilled workers have similar levels. As with education, the mechanism behind the heterogeneous elasticities is unclear, but it is highly likely to be a combination of different bargaining power levels and wage profiles.

Industry The results by industry in Table 22 showcase the common trends aforemen-

tioned: industries with a higher share of educated, high-skilled professionals have weaker elasticities. In manufacturing, trade and transportation, where more than 50% of the sample lies, the elasticities range from -0.04 to -0.07, reasonably in line with aggregate results. The biggest outlier is the construction industry, with a coefficient of -0.27, which, considering the composition of its workers, is in line with the previous results on education and occupation heterogeneity.

Labor market concentration Lastly, Table 23 presents the estimation results by level of labor market concentration. A low concentration market is one where the market-level HHI is below the national median, computed yearly. In line with recent research relating labor market power to monopsony power, the results claim that more concentrated markets have a lower elasticity of wages to other firms' pay policies. This relates to the notion that in highly concentrated markets the outside options truly outside of a firm are much fewer, implying lower competition for workers and thus granting firms a level of monopsony power over workers (Jarosch, Nimczik, and Sorkin 2024). Thus, the quit/renegotiation channel is much weaker in more concentrated markets because the chances of getting better offers are much lower (Berger et al. 2024).

TABLE 21. Elasticity of wages to other firms' pay premia, by occupation group

	(1) Managers	(2) Skilled Profs.	(3) Techn. & Ass. Profs	(4) Clerical Supp.	(5) Service & Sales	(6) Craft Trades	(7) Elementary Occupations
$\overline{\zeta}_{-fgt}$	-0.3106	-0.0409***	-0.0763***	-0.0713***	-0.1805***	-0.4097***	-0.3707***
	(0.2007)	(0.0147)	(0.0115)	(0.0119)	(0.0162)	(0.0112)	(0.0151)
Worker FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Firm FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Market-Year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Weighted?	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	14,170	1,623,555	1,166,348	1,906,094	3,395,158	1,811,558	1,193,078

Note: Coefficients from separate regressions for each occupation group. Split-sample refers to incidental parameter bias adjusment as in Dhaene and Jochmans (2015). See Section 2.3.3 for details. Weighted refers to current employment weights used in the computation of the leave-out average of firm effects. Standard errors are clustered at firm × year level in parenthesis.
*** Denotes statistical significance at the 0.01 level.

TABLE 22. Elasticity of wages to other firms' pay premia, by industry

	(1) Manufacturing	(2) Construction	(3) Trade & Transport	(4) Accom. & Restaurants	(5) High VA Services	(6) Other Services
$\hat{\gamma}$	-0.0446***	-0.2709***	-0.0741***	-0.0459***	0.0090	-0.0031
	(0.0074)	(0.0132)	(0.0107)	(0.0138)	(0.0156)	(0.0102)
Worker FE	Yes	Yes	Yes	Yes	Yes	Yes
Firm FE	Yes	Yes	Yes	Yes	Yes	Yes
Market-Year FE	Yes	Yes	Yes	Yes	Yes	Yes
Weighted?	Yes	Yes	Yes	Yes	Yes	Yes
Observations	2,910,280	833,489	3,356,852	1,033,991	1,445,744	3,024,315

Note: Coefficients from separate regressions for each industry or group of industries. High VA services = Information and Communications; Financial Services; Real Estate Activities; Professional, Scientific and Technical Activities. Split-sample refers to incidental parameter bias adjusment as in Dhaene and Jochmans (2015). See Section 2.3.3 for details. Weighted refers to current employment weights used in the computation of the leave-out average of firm effects. Standard errors are clustered at firm × year level.
*** Denotes statistical significance at the 0.01 level.

TABLE 23. Elasticity of wages to other firms' pay premia, by labor market concentration

	Low concentration		High concentration	
$\hat{\gamma}$	-0.2004***	-0.3258***	-0.1515***	-0.1118***
	(0.0109)	(0.0100)	(0.0057)	(0.0265)
Worker FE	✓	✓	✓	✓
Firm FE	✓	✓	✓	✓
Market-Year FE	✓	✓	✓	✓
Split-Sample?	–	✓	–	✓
Weighted?	✓	✓	✓	✓
Observations	9,102,954	9,102,954	5,766,811	5,766,811

Note: Low concentration = 1 when peer group belongs to a labor market with HHI below the national median. Split-sample refers to incidental parameter bias adjusment as in Dhaene and Jochmans (2015). See Section 2.3.3 for details. Weighted refers to current employment weights used in the computation of the leave-out average of firm effects. Standard errors are clustered at firm × year level.
*** Denotes statistical significance at the 0.01 level.

F. Theoretical motivation

F.1. Wage backloading through search and renegotiation

The models that motivate this work are derivative of Postel-Vinay and Robin (2002) and Cahuc, Postel-Vinay, and Robin (2006). In essence, these present labor markets where firms post vacancies and workers search on the job for new offers. Upon meeting an offer, workers decide whether to take that offer by taking it to their employer, which triggers a renegotiation phase that mimics a second-price auction. In the end, workers either move to a new firm that offers a better contract or stay in their current firm with a renegotiated wage. Importantly, wages are initially set through Nash bargaining, granting workers some share of the match surplus.

To better approximate the reduced form model employed in this paper, consider a variation of this framework with a finite number of firms. This setting, borrowed from Jarosch, Nimczik, and Sorkin (2024) and Berger et al. (2024), generates firms with different discrete sizes, with workers being more likely to meet larger firms during search. Furthermore, consider that employees partially direct their search while on the job, as they do not consider positions in their current employer. These ingredients create within- and between-market bargaining positions and outside options. Wages depend on a weighted average of the match surplus of working at each firm in the market.

Wage equation The wage equation from Berger et al. (2024) is a good motivating example of the dynamics that this paper wants to capture[3]:

$$w_i(\sigma) = \sigma z_i + (1-\sigma)b + \beta \Bigg[(1-\sigma)\theta \sum_k \lambda_{uk} \max\{S_k, 0\} - \sum_{k \neq i} \lambda_{ik} \max\{0, \min\{(1-\sigma)\theta(S_k - S_i), (S_k - S_i)\} + (1-\sigma)S_i\} \Bigg] \tag{39}$$

where $w_i(\sigma)$ is the wage of worker in firm i as a function of the match surplus share, σ; z_i is the productivity of the firm she works for; b is the flow value of unemployment; θ is the bargaining power of the worker; λ_{ik} is the meeting rate between a worker in firm i and a vacancy in firm k; S_i is the match surplus of working in firm i.

In this setting, workers get:

a. Part of the match productivity: σz_i;
b. A share of their outside option: $(1-\sigma)\left(b + \beta\theta \sum_k \lambda_{uk} \max\{S_k, 0\}\right)$;
c. A (negative) compensating differential from quits/renegotiation: $-\beta \sum_{k \neq i} \lambda_{ik} \max\{0, \min\{(1-\sigma)\theta(S_k - S_i), (S_k - S_i)\} + (1-\sigma)S_i\}$.

[3] For exposition purposes, this version of the equation ignores non-wage amenities

This shows that the wage policies of other firms can directly influence wages through channels b) and c). First, workers, once they find a job, negotiate their wage in a Nash bargaining setting and get some share of the match surplus; second, due to positive future wage increases stemming from on-the-job search and bargaining, firms tend to downgrade wages at earlier tenures. This steepness of the wage schedule generates a form of monopsony power and generally depresses wages. Empirical exercises that estimate the impact of outside options on wages generally fail to disentangle these mechanisms. This paper addresses both channels from a reduced-form approach, highlighting the importance of the outside option and the quit/renegotiation channels in determining wages.

F.2. Human capital acquisition and learning

Gualdani et al. (2026) provide a complementary mechanism: employment develops human capital and reveals initially unknown worker ability. Heterogeneous firms compete à la Bertrand and differ in their production, training, and information technologies. Publicly observed output supplies signals about ability.

Wage equation Let j be the employing firm and k the best alternative, ranked by lifetime wage value. With a common exogenous labor-market exit probability σ, Proposition 1 of Gualdani et al. (2026) can be written as

$$w_{it} = \underbrace{y_k(s_{it})}_{\text{current outside option}} + \underbrace{\delta(1-\sigma)\big[C_k(k;s_{it}) - C_k(j;s_{it})\big]}_{\text{human capital and learning differential}} . \tag{40}$$

Here, s_{it} collects human capital, worker efficiency, employment history, common beliefs about ability, and current productivity shocks; $y_k(s_{it})$ is expected current output at k; and δ is the discount factor. $C_k(d;s_{it})$ denotes expected next-period continuation match surplus for the worker and firm k, conditional on employment at d today: the sum of the worker's lifetime wage value and firm k's lifetime profit value. Both continuation terms therefore concern firm k.

The differential is negative when employment at j offers more valuable human capital or information. In the paper's two-firm case with identical human capital technologies and exit risks, Proposition 3 links this sign to j providing a more informative ability signal. Workers accept lower current pay for these future returns.

To connect this mechanism to alternative employers, consider an illustrative case in which a specialist job at k becomes more productive for workers whose ability is revealed to be high. If workers with their current information would still choose another job at k, its current output $y_k(s_{it})$ need not rise. However, information acquired at j can now be more valuable for future job assignment. If this raises $C_k(j;s_{it})$ more than $C_k(k;s_{it})$,

the employing firm can pay a lower current wage. More generally, holding the employer ranking fixed, better alternatives reduce current pay when they increase this continuation-value gap enough to outweigh any rise in current outside productivity. This is a conditional implication of the wage equation; better learning opportunities at the alternative employer itself can instead raise the compensation required to retain the worker.

For the empirical analysis, this mechanism motivates examining career stage alongside tenure: the value of information depends on remaining uncertainty about ability, while tenure measures time with the current employer. The leave-out average of other firms' pay premia does not directly measure learning opportunities or their future returns. A negative conditional relationship with current wages is therefore consistent with this mechanism when better alternatives increase the value of learning at the current job, but does not by itself identify that channel.

G. Variance decomposition bias correction

Recent studies have called attention to the need for bias correction in models with high-dimensional fixed effects due to the incidental parameter bias (Kline, Saggio, and Sølvsten 2020; Bonhomme et al. 2023). The authors have claimed this to be a problem stemming from limited mobility in the worker-firm network. Because worker and firm fixed effects are identified via firm-to-firm moves, lower mobility induces a high uncertainty in the estimation of the fixed effects, which leads to larger worker and firm fixed effect variances, to the detriment of covariances.

The most used bias correction techniques for the variance are those in Andrews et al. (2008) and Kline, Saggio, and Sølvsten (2020). Replicating the formulas in Bonhomme et al. (2023), consider the design matrix A, which represents the worker-firm-market graph of the network. Consider also $\xi = (\alpha, \zeta, \theta)$. Then, an estimator for ξ is

$$\hat{\xi} = (A'A)^{-1}A'W \tag{41}$$

The estimator for the variance components is, for some matrix Q:

$$\hat{V}_Q = \hat{\xi}' Q \hat{\xi} \tag{42}$$

The bias in $\hat{V}_Q$ can be written as

$$\mathbb{E}\left[\varepsilon' A(A'A)^{-1}Q(A'A)^{-1}A'\varepsilon\right] \tag{43}$$

The bias correction in Andrews et al. (2008) is constructed under the homoskedasticity assumption as

$$\hat{V}_Q^{HO} = \hat{V}_Q - \hat{\sigma}^2 \text{Trace}\left((A'A)^{-1}Q\right) \tag{44}$$

The correction in Kline, Saggio, and Sølvsten (2020) relaxes the homoskedasticity assumption and is given by

$$\hat{V}_Q^{HO} = \hat{V}_Q - \text{Trace}\left(A(A'A)^{-1}Q(A'A)^{-1}A'\hat{\Omega}A\right), \tag{45}$$

where $\hat{\Omega}$ is a diagonal matrix with terms $\hat{\sigma} = W_{it}\left(W_{it} - \hat{\alpha}_i^{-(i,t)} - \hat{\zeta}_{f(i,t)}^{-(i,t)} - \hat{\theta}_{m(i,t)}^{-(i,t)}\right)$. $\hat{\sigma}$ is obtained from a leave-out jackknife procedure, where each observation (i, t) is removed from the sample sequentially. Kline, Saggio, and Sølvsten (2020) provide a simple way to avoid reestimating the model for millions of iterations.

Figure 13 shows the results from the estimator without bias correction (FE), with Andrews et al. (2008)'s correction (HO) and with Kline, Saggio, and Sølvsten (2020)'s correction

(HE).

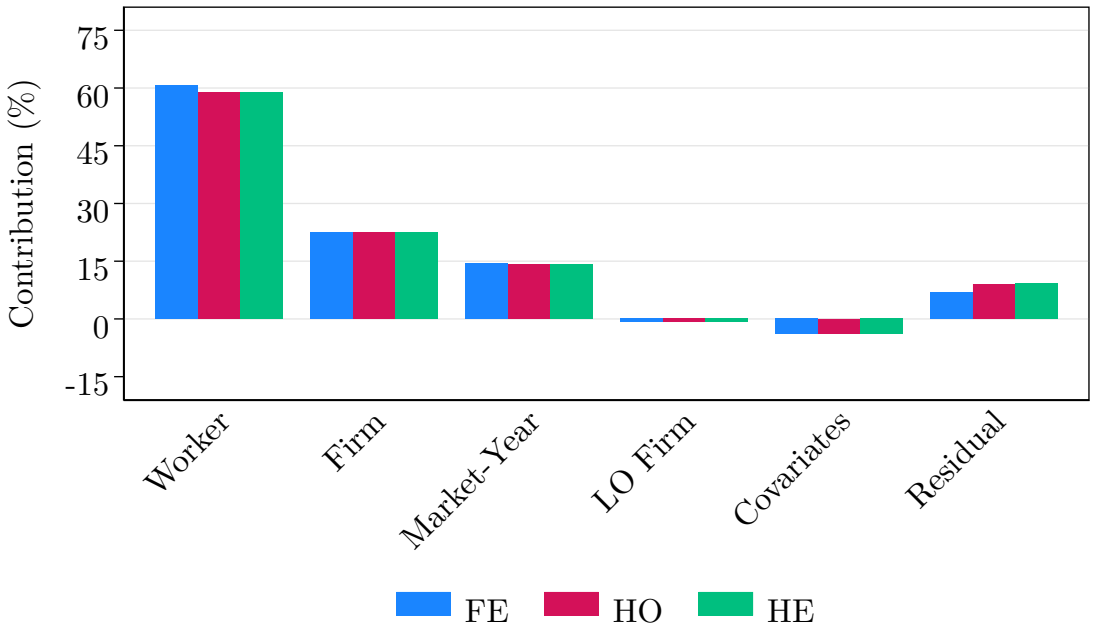


FIGURE 13. Firm Effects comparison: AKM versus NLS estimation.

Note: FE: non-linear least squares estimation without any bias adjustment correction for the variance or the coefficients; HO: bias in variance corrected according to Andrews et al. (2008); HE: bias in variance corrected according to Kline, Saggio, and Sølvsten (2020). The contribution of each component is calculated according to (18). Unlike the figures and tables in the main text, worker fixed effects are not adjusted for the linear effect of age and gender, which explains the differences between FE and the results in Figure 5.

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